I

VOLUME I · REVISION NOTES

Notes In Prosthodontics

A complete Diploma revision companion · Integrated edition

Fourteen clinically organised units, with original Diploma notes retained
Integrated F Pros explanations and clinical decision-making guidance
Applied anatomy, materials, removable and fixed prosthodontics
Implants, tooth wear, occlusion, aesthetics and digital workflows
High-yield numbers, safe-practice principles and referral decisions

Mapped to the Diploma in Prosthodontics – Learning Outcomes (August 2026); informed by NHS England Level 2 complexity and restorative dentistry standards.

For candidates preparing for the written and structured oral examinations

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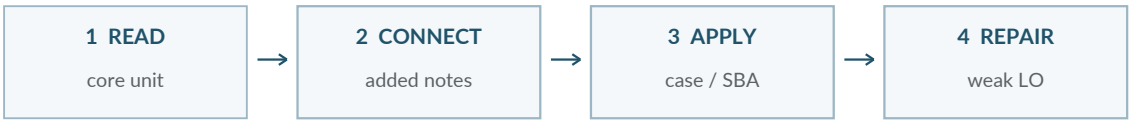
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The revision companion

One integrated volume, designed to move between the original Diploma learning units, added clinical explanations and the published learning outcomes. Use it as a study guide and a reference during revision.

I	**The 14 Diploma units** Original content retained	The core that teaches the material. Anatomy, materials, assessment, removable and fixed prosthodontics, occlusion, implants, maintenance, governance and high-yield numbers.
II	**Integrated companion notes** Additional reading inserted by unit	Selected Fellowship material consolidated into the appropriate Diploma units. Extra clinical context explains decision-making, prevention, imaging, tooth wear, aesthetics and digital workflow.
III	**Syllabus and source maps** Learning-outcome alignment	The August 2026 Diploma learning-outcome domains guide the depth of revision. The integration and reference map on the next page shows where the additional material belongs.

A simple study cycle



Read the core unit first. Follow immediately with its integrated notes, then test whether you can explain the diagnosis, first safe action, alternatives, consent, follow-up and referral. Revisit the associated learning-outcome code when an answer is weak.

Diploma scope at a glance

Applied knowledge	**1.1–1.11**	Basic biological sciences
Clinical judgement	**2a.1–2a.21**	Assessment and diagnosis
Clinical judgement	**2b.1–2b.12**	Treatment planning and management
Clinical judgement	**2c.1–2c.27**	Prosthodontic-specific planning
Good practice	**3a.1–3a.12**	Clinical governance
Good practice	**3b.1–3b.5**	Communication

Scope note. These are Level 2 / enhanced-skills outcomes, not the specialist Fellowship blueprint. Advanced specialist interventions are not presented as Diploma requirements; know when to refer. The original notebook describes 150 SBAs in 180 minutes, but the supplied learning outcomes independently specify only the written and SOE components. Confirm the current examination instructions.

EDITORIAL INTEGRATION MAP

What has been combined

F Pros chapters redistributed to the appropriate Diploma units.

FELLOWSHIP SOURCE	DIPLOMA HOME	WHAT WAS RETAINED
F 1: Examination, diagnosis	Unit 1 / 4 / 5	Anatomy, medical modifiers, reasoning
F 2: Clinical investigations	Unit 5	Imaging and test limitations
F 3: Treatment strategies	Unit 5	Planning, consent and phasing
F 4: Health promotion	Unit 2	Disease prevention and maintenance
F 5: Fixed prosthodontics	Units 3 / 8 / 12	Materials, design and review
F 6: Removable prosthodontics	Units 6 / 7 / 12	RPD, complete dentures and outcomes
F 7: Implant prosthodontics	Unit 11	Planning, disease and referral only
F 8: TMD	Unit 9	Reversible care and red flags
F 9: Erosive tooth wear	Unit 10	Wear monitoring and additive care
F 10: Aesthetic dentistry	Unit 8	Conservative choices
F 11: Materials science	Unit 3	Property-led material selection
F 12: Dental traumatology	Unit 10	Recognition and consequences
F 13: Digital dentistry	Unit 11	Accuracy and workflow limits
F crash notebook	Unit 14	Decision traps only

REFERENCE HIERARCHY

1. **Diploma in Prosthodontics – Learning Outcomes** (Aug 2026): scope and domains. 2. **NHS England Clinical standard for restorative dentistry** (v2, 28 Oct 2022): complexity and referral (especially the prosthodontic decision map, p. 8). 3. **Accreditation of Performers of Level 2 complexity care** (May 2019): general accreditation context; its supplied appendices do not set a bespoke prosthodontic syllabus. 4. **GDC Prosthodontics Specialty Training Curriculum** (2024): specialist background only, not a substitute for Diploma requirements.

Learning outcome code ranges follow the supplied August 2026 Diploma document.

UNIT 1

Applied anatomy

LO 1.1, 1.5 | Question Bank Block A (Q1–90) | Block N (Q1001–1090) | Crash notebook Schematics 1–4 | Number card 1

IN ONE MINUTE

All four muscles of mastication are first-arch, supplied by V3 and fed by the maxillary artery; only the lateral pterygoid opens the mouth.

Elevators: masseter, temporalis, medial pterygoid. **Main retractor:** posterior temporalis.

V3: anterior division motor except buccal; posterior division sensory except the nerve to mylohyoid.

Denture borders are set by muscles: modiolus, masseter, mylohyoid, and the coronoid process.

All four muscles of mastication develop from the first pharyngeal arch, are supplied by the mandibular division of the trigeminal nerve (V3), and receive blood from branches of the maxillary artery. The lateral pterygoid is the only one that opens the mouth.

1.1 Muscles of mastication

MUSCLE	ORIGIN	INSERTION	MAIN ACTION	NERVE	BLOOD SUPPLY
Masseter (superficial)	Zygomatic process of maxilla and anterior two-thirds of lower border of zygomatic arch	Angle and lower lateral surface of ramus	Elevation; slight protrusion	Masseteric nerve (anterior division of V3), passes through the mandibular (sigmoid) notch	Masseteric artery (2nd part of maxillary); transverse facial and facial arteries
Masseter (deep)	Posterior third and deep surface of zygomatic arch	Upper lateral ramus and coronoid process	Elevation; posterior fibres retrude	As above	As above
Temporalis	Floor of temporal fossa (below inferior temporal line) and deep surface of temporal fascia	Coronoid process (apex, medial surface) and anterior border of ramus to retromolar fossa	Elevation; posterior (horizontal) fibres retrude; main positional muscle	Deep temporal nerves (anterior division of V3)	Anterior and posterior deep temporal arteries (2nd part of maxillary); middle temporal (superficial temporal)
Medial pterygoid (deep head)	Medial surface of lateral pterygoid plate; pyramidal process of palatine bone	Medial surface of ramus and angle (pterygoid tuberosity)	Elevation; protrusion with lateral pterygoid; contralateral side-shift	Nerve to medial pterygoid (from main trunk of V3, before division)	Pterygoid branches of maxillary artery
Medial pterygoid (superficial head)	Maxillary tuberosity and pyramidal process of palatine	As above	As above	As above	As above
Lateral pterygoid (superior head)	Infratemporal surface and crest of greater wing of sphenoid	Articular disc and capsule; pterygoid fovea	Stabilises condyle and disc during closing (power stroke)	Nerve to lateral pterygoid (anterior division of V3, often with buccal nerve)	Pterygoid branches of maxillary artery
Lateral pterygoid (inferior head)	Lateral surface of lateral pterygoid plate	Pterygoid fovea on neck of condyle	Protrusion, opening (depression); unilateral contraction moves	As above	As above

		mandible to the opposite side

EXAM TRAPS

- The medial pterygoid nerve comes off the main trunk of V3; it also carries fibres through the otic ganglion to tensor veli palatini and tensor tympani.
- The masseter and medial pterygoid form a sling around the angle of the mandible.
- Disc displacement and TMJ clicking and locking are most associated with the lateral pterygoid (superior head attaches to the disc).
- Masseteric hypertrophy is a sign of bruxism; temporalis is the muscle most often palpably tender in tension-type headache linked to TMD.

1.2 Mandibular nerve (V3) branches

- **Main trunk:** nervus spinosus (meningeal branch) and nerve to medial pterygoid.
- **Anterior division (mainly motor):** masseteric, deep temporal, nerve to lateral pterygoid, and the buccal (long buccal) nerve — the only sensory branch of the anterior division.
- **Posterior division (mainly sensory):** auriculotemporal (embraces the middle meningeal artery; secretomotor to parotid), lingual (joined by chorda tympani), inferior alveolar (gives the nerve to mylohyoid, motor to mylohyoid and anterior belly of digastric; ends as mental and incisive nerves).

1.3 Maxillary artery

From the external carotid, behind the condylar neck.

PART	POSITION	BRANCHES
1st (mandibular)	Behind neck of condyle	Deep auricular, anterior tympanic, middle meningeal (through foramen spinosum), accessory meningeal, inferior alveolar
2nd (pterygoid)	Related to lower head of lateral pterygoid	Deep temporal, pterygoid, masseteric, buccal
3rd (pterygopalatine)	In pterygopalatine fossa	Posterior superior alveolar, infraorbital, descending (greater and lesser) palatine, artery of pterygoid canal, pharyngeal, sphenopalatine

Venous drainage of the muscles is to the pterygoid venous plexus, which communicates with the cavernous sinus via emissary veins — the route for intracranial spread of infection. Lymph from the muscles drains to deep cervical nodes.

1.4 Other muscles that move the mandible or shape dentures

MUSCLE	NERVE	RELEVANCE
Anterior belly of digastric	Nerve to mylohyoid (V3)	Depresses mandible; attaches at digastric fossa
Posterior belly of digastric	Facial nerve (VII)	Elevates hyoid
Mylohyoid	Nerve to mylohyoid (V3)	Floor of mouth; limits lingual flange of lower denture in the molar region
Geniohyoid	C1 via hypoglossal nerve	Depresses mandible, pulls hyoid forward
Stylohyoid	Facial nerve	Elevates and retracts hyoid
Buccinator	Facial nerve (buccal branches)	Lies over the buccal shelf; keeps food on the occlusal table; shapes the buccal polished surface

Orbicularis oris, modiolus muscles	Facial nerve	Determine lip support, neutral zone and the labial flange
Genioglossus	Hypoglossal (XII)	Lingual frenum; limits anterior lingual flange
Palatoglossus	Vagus via pharyngeal plexus	Only tongue muscle not supplied by XII
Tensor veli palatini	Nerve to medial pterygoid (V3)	Only soft palate muscle not supplied by the pharyngeal plexus

Gag reflex: afferent glossopharyngeal (IX), efferent vagus (X).

1.5 Temporomandibular joint

- **Type:** synovial, ginglymoarthrodial (hinge and gliding) joint; articular surfaces covered by fibrocartilage (not hyaline), reflecting its intramembranous origin.
- **Disc:** biconcave and avascular centrally; posteriorly the bilaminar zone (retrodiscal tissue) is vascular and innervated — the source of pain in disc displacement.
- **Compartments:** lower = rotation (first 20–25 mm of opening); upper = translation.
- **Ligaments:** lateral (temporomandibular) ligament is the true stabiliser and limits posterior and inferior movement; sphenomandibular (from spine of sphenoid to lingula) and stylomandibular are accessory.
- **Nerve supply:** auriculotemporal (main), masseteric and deep temporal (Hilton's law).
- **Blood:** superficial temporal and maxillary arteries.

1.6 Denture-bearing anatomy

ARCH	PRIMARY SUPPORT	RELIEF AREAS	BORDER LIMITS
Maxilla	Horizontal hard palate and residual ridge crest (posterior)	Incisive papilla (nasopalatine nerve), midline raphe, torus palatinus	Labial and buccal frena; buccal vestibule; coronoid process limits distobuccal flange width; hamular notches; vibrating line (post-dam)
Mandible	Buccal shelf (between external oblique ridge and residual ridge; cortical bone, at right angles to occlusal load)	Mental foramen (on crest in severe resorption), genial tubercles, sharp mylohyoid ridge, knife-edge crest	Labial and buccal frena; masseteric notch (distobuccal); retromolar pad (cover at least the anterior half to two-thirds); mylohyoid (lingual flange slopes towards tongue in molar region); genioglossus and lingual frenum

- The post-dam lies between anterior and posterior vibrating lines; fovea palatini are a guide to its position.
- The hamular (pterygomaxillary) notch marks the posterior limit of the tuberosity and the post-dam's lateral extent; the pterygomandibular raphe runs from hamulus to retromolar region.
- Parotid duct opens opposite the upper second molar; it can be compressed by an over-extended buccal flange.

1.7 Tissue spaces and spread of infection

- **Mylohyoid line rule:** infection from lower teeth whose apices lie below the mylohyoid attachment (second and third molars) spreads to the submandibular space; teeth above it (incisors to first molar) spread to the sublingual space.
- **Ludwig's angina:** bilateral involvement of submandibular, sublingual and submental spaces; airway emergency.
- **Canine space:** upper canine infection above levator anguli oris enters it; risk of cavernous sinus thrombosis via the angular and ophthalmic veins.
- **Pterygomandibular space:** contains the inferior alveolar and lingual nerves and vessels; the target space of the inferior alveolar block and a route to the lateral pharyngeal space.

- **Submasseteric and buccal spaces:** present with trismus and cheek swelling.

1.8 Salivary glands

GLAND	SECRETION	DUCT OPENING	PARASYMPATHETIC ROUTE
Parotid	Serous	Opposite upper second molar (Stensen's)	IX → lesser petrosal → otic ganglion → auriculotemporal
Submandibular	Mixed, mainly serous; about 65–70% of unstimulated flow	Sublingual papilla (Wharton's)	VII → chorda tympani → lingual → submandibular ganglion
Sublingual	Mainly mucous	Multiple ducts of Rivinus on sublingual fold	As submandibular

Parotid contributes most of stimulated saliva.

Examiner traps — Unit 1

1. The nerve to medial pterygoid arises from the main trunk, not a division.
2. The lateral pterygoid inserts into the pterygoid fovea, disc and capsule — not the coronoid.
3. Buccinator and the posterior belly of digastric are facial nerve; the buccal branch of V3 is sensory.
4. Lower molar infection below the mylohyoid line goes to the submandibular space.

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Clinical anatomy in decision-making

Source bridge · F Pros Chapter 1

DIPLOMA LEARNING OUTCOMES 1.1, 1.5, 1.11 · 2a.5–2a.6

CORE PRINCIPLE

Translate anatomy into risk, denture extension and examination findings—not an isolated list of muscle names.

01 Functional anatomy at the chairside

Map the muscles and frena to the denture borders: lip and cheek muscles shape the polished surface; the mylohyoid, retromylohyoid area and tongue control lower lingual extension. An apparently retentive denture may still be unstable if its contours conflict with muscle movement.

02 Structures that alter the plan

In an atrophic mandible, consider the proximity of the mental nerve to the denture-bearing surface. In the maxilla, examine palatal form, tuberosities and soft-palate movement before deciding the posterior palatal seal or a palatal connector.

03 Risk and referral

Anatomical complexity becomes clinically relevant when it changes radiographic selection, impression design or safety. Spreading infection with airway concern, complex soft-tissue pathology or unexplained neuropathy requires escalation; detailed surgery is outside this Diploma supplement.

04 One-best-answer cue

A short mandibular lingual flange may lose the seal; excessive extension can dislodge the denture during tongue action. Diagnose the functional border before changing the occlusion or adding adhesive.

SCOPE / GUIDANCE

Level 2: recognition, examination, prosthesis design and timely referral. No specialist anatomical surgery is implied.

Source material: F Pros pp. 8–10 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 2

Physiology and pathology

LO 1.2–1.4, 1.8, 1.10–1.11, 2a.7–2a.8 | Question Bank Block B (Q91–160) | Block O (Q1091–1160) | Crash notebook Schematic 5 | Number card 1

IN ONE MINUTE

Frequency of sugar and acid, not the total amount, drives caries and erosion; saliva both retains dentures and protects teeth.

Critical pH: enamel 5.5, root 6.2–6.7, fluorapatite 4.5.

After tooth loss the maxilla resorbs centripetally, the mandible centrifugally, and the anterior mandible about four times faster.

An ulcer that persists 3 weeks after removing the cause is an urgent referral; erythroplakia carries the highest malignant risk.

After tooth loss the residual ridge resorbs fastest in the first 6–12 months and continues for life; the anterior mandible loses about four times as much height as the anterior maxilla in the long term (Tallgren).

2.1 Saliva

- **Volume:** about 0.5–1.5 L/day. Unstimulated flow about 0.3–0.4 mL/min. **Hyposalivation** = unstimulated below 0.1 mL/min or stimulated below about 0.5–0.7 mL/min.
- **Functions:** lubrication, buffering (bicarbonate main buffer in stimulated saliva), remineralisation, antimicrobial (lysozyme, lactoferrin, IgA, histatins), taste, bolus formation, denture retention.
- **Denture retention** depends on a thin film of saliva: adhesion (saliva to mucosa and denture), cohesion (within saliva), interfacial surface tension, viscosity, and atmospheric pressure via the border (peripheral) seal. Thin watery saliva gives better retention than thick ropy saliva.
- **Common causes of dry mouth:** drugs (antidepressants, antihistamines, antimuscarinics, diuretics, antipsychotics), Sjögren's syndrome, head and neck radiotherapy, poorly controlled diabetes, dehydration, anxiety, mouth breathing.
- **Management:** treat cause, sugar-free stimulants, saliva substitutes (mucin- or carboxymethylcellulose-based; avoid acidic substitutes in dentate patients), pilocarpine (contraindicated in asthma, glaucoma), high-fluoride toothpaste.

2.2 Mastication and neuromuscular control

- Periodontal mechanoreceptors give fine tactile and load feedback; they are lost in the edentulous patient, so complete denture wearers depend on mucosal and TMJ receptors. Implants give "osseoperception", less sensitive than a natural tooth.
- Complete denture wearers achieve roughly **one-fifth to one-sixth** of the bite force and masticatory efficiency of dentate people.
- **Shortened dental arch (Käyser):** an arch of premolars and anterior teeth (about 3–5 occlusal units) usually gives adequate function in older adults; replacing missing molars with a free-end RPD often adds little.
- **WHO goal:** retention of at least 20 functional teeth throughout life.
- Ability to adapt to new dentures falls with age, neuromuscular disease (Parkinson's), dementia and stroke: favour copy (replica) techniques.

2.3 Consequences of tooth loss

- **Pattern:** maxilla resorbs upwards and inwards (centripetal), making the upper arch smaller; mandible resorbs downwards and outwards (centrifugal), making it wider. Result: an apparent Class III relationship and posterior crossbite in long-term edentulous patients.
- **Local factors:** drifting, tilting and over-eruption of opposing teeth (reducing interocclusal space), loss of occlusal vertical dimension, loss of lip and cheek support.
- **Resorption factors:** anatomical (bone volume and density), metabolic (osteoporosis, age, sex), functional (parafunction),

prosthetic (ill-fitting dentures, continuous wear, inadequate support).

- **Ridge preservation strategies:** retain roots as overdenture abutments, implants, immediate dentures to protect the socket, remove dentures at night.

2.4 Caries

- **Critical pH:** enamel (hydroxyapatite) about 5.5; root dentine and cementum about 6.2–6.7; fluorapatite about 4.5.
- **Stephan curve:** pH falls within minutes of sugar intake and recovers in 30–60 minutes; frequency of intake matters more than quantity.
- **Microbiology:** Streptococcus mutans (initiation; acidogenic, aciduric, extracellular polysaccharides), Lactobacillus (progression in cavitated lesions), Actinomyces (root caries). Ecological plaque hypothesis: environmental change (frequent sugar, low saliva) selects for acid-producing species.
- **Enamel lesion zones (deep to surface):** translucent zone, dark zone, body of lesion (largest demineralisation), surface zone (relatively intact).
- **Dentine lesion zones:** infected (soft, bacteria-laden, remove) and affected (demineralised, remineralisable, can be left in selective removal).
- **Delivering Better Oral Health (2021):** adults 1450 ppm F toothpaste, spit don't rinse; high-risk adults 5000 ppm (16 and over) or 2800 ppm (10 and over); fluoride varnish 22,600 ppm at least twice yearly for those at higher risk.
- **Root caries risk** rises with recession, hyposalivation, partial dentures (plaque retention around clasps and saddles), poor dexterity, and frequent sugar.

2.5 Non-carious tooth surface loss

TYPE	MECHANISM	TYPICAL APPEARANCE
Erosion (extrinsic)	Dietary acids: carbonated drinks, citrus, wine, vinegar	Labial surfaces of upper incisors, cupped cusps, smooth silky enamel, restorations standing proud
Erosion (intrinsic)	Gastric acid: GORD, bulimia, alcoholism, pregnancy vomiting	Palatal surfaces of upper anteriors (perimylolysis), occlusal surfaces of lower molars
Attrition	Tooth-to-tooth contact (bruxism)	Flat shiny facets that match in opposing teeth, fractured cusps, hypertrophic masseters
Abrasion	Foreign object (toothbrush, pipe, hair grips)	Sharp-edged V-shaped cervical notches, worse on the side opposite the brushing hand
Abfraction (debated)	Flexural stress concentration at the cervical margin	Wedge-shaped cervical lesions with sharp angles, may be subgingival

- **Bulimia signs:** palatal erosion of upper anteriors, parotid enlargement, Russell's sign (knuckle calluses), dentine hypersensitivity. Patients may be secretive; approach with empathy and liaise with GMP.
- Compensatory alveolar growth often maintains OVD in slow wear, so there may be no interocclusal space — the reason for the Dahl approach (Section 10).
- **TSL indices:** Smith and Knight Tooth Wear Index (scores 0–4 per surface), BEWE (Basic Erosive Wear Examination, 0–3 per sextant, sums guide management).

2.6 Oral mucosa and dentures

CONDITION	KEY FACTS	MANAGEMENT
Denture stomatitis	Chronic erythematous candidosis under the upper denture; Newton type I (pinpoint hyperaemia), II (diffuse erythema), III (papillary hyperplasia of the central palate)	Denture hygiene, leave out at night, soak in hypochlorite (acrylic only) or chlorhexidine, correct fit; antifungal (miconazole — interacts with warfarin and statins; or nystatin) if persistent

Angular cheilitis	Candida and/or Staphylococcus aureus; predisposed by reduced OVD, haematinic deficiency, diabetes, immunosuppression	Treat denture source, check FBC/haematinics/glucose; miconazole cream (also active against S. aureus) or fusidic acid
Denture-induced hyperplasia	Fibrous overgrowth in the sulcus from an over-extended or resorbed-denture flange	Trim flange, tissue conditioner; excise and send for histology if it does not resolve
Traumatic ulcer	Under an over-extended flange or sharp edge	Adjust; review in 2 weeks; persistent ulcer beyond 3 weeks → urgent suspected cancer referral
Flabby ridge	Fibrous replacement of bone in the anterior maxilla, often opposing lower natural anterior teeth (combination syndrome)	Mucostatic or selective-pressure impression (windowed tray)

2.7 Potentially malignant disorders and oral cancer

- **Erythroplakia** has the highest malignant transformation risk; leukoplakia (non-homogeneous, speckled, floor of mouth, lateral tongue) higher risk than homogeneous.
- **Highest-risk sites** for squamous cell carcinoma: lateral border of the tongue and floor of mouth.
- **Risk factors:** tobacco and alcohol (synergistic), areca nut, HPV (oropharynx), immunosuppression, previous oral cancer.
- **NICE NG12:** unexplained oral ulceration over 3 weeks, persistent lump in the mouth or lip, red or red-and-white patch → urgent referral (suspected cancer pathway).

2.8 Ageing and systemic links

- **Ageing teeth:** secondary dentine, smaller pulp chambers (harder access, lower sensibility responses), brittle dentine, darker shade.
- **Haematinic deficiency** (iron, B12, folate): glossitis, angular cheilitis, recurrent ulcers, burning sensation.
- **Diabetes:** periodontitis (bidirectional), candidosis, xerostomia, delayed healing, burning mouth.
- **Osteoporosis:** generally not a contraindication to implants; drug treatment (antiresorptives) is the bigger issue (MRONJ, Section 4).

Examiner traps — Unit 2

1. Root surfaces demineralise at a higher pH than enamel, not a lower one.
2. Newton type III denture stomatitis is papillary hyperplasia.
3. Miconazole, even as a gel, interacts with warfarin and statins.
4. Restorations standing proud of the tooth point to erosion, not attrition.

Previous
1 · Applied anatomy

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3 · Dental materials science

Prevention as part of prosthodontic care

F Pros Chapter 4 → Diploma Unit 2

DIPLOMA LEARNING OUTCOMES 1.4, 1.8 · 2b.3–2b.5 · 2c.1

CORE PRINCIPLE

Every definitive restoration or prosthesis depends on disease control and realistic maintenance.

01 High-yield preventive advice

The source F Pros notes cite Delivering Better Oral Health v4: brush twice daily with 1,350–1,500 ppm fluoride toothpaste, then spit without rinsing. Higher-fluoride products and varnish are selected by caries risk, age, medical history and the current prescribing guidance.

02 A workable sequence

Control sugar and acid frequency, tobacco use, plaque accumulation and active periodontal disease before definitive treatment. The F Pros notes summarise periodontal care as oral hygiene/risk-factor control → subgingival instrumentation → surgery where necessary → supportive periodontal care.

03 Denture-specific prevention

Take dentures out overnight and clean them with non-abrasive products. Sodium hypochlorite can damage cobalt-chromium; avoid it for metal frameworks. Manage denture stomatitis by addressing fit and hygiene as well as candidal infection.

04 Patient factors change designs

Xerostomia can impair retention and increase caries and candidosis. When cleaning or attendance will be unreliable, favour hygienic, maintainable designs rather than elaborate fixed work that the patient cannot support.

05 SBA traps

Miconazole interacts with warfarin; check current prescribing advice. Deep caries may favour selective caries removal rather than complete removal over the pulp. Do not assume every missing tooth needs replacement.

SCOPE / GUIDANCE

Prescribing and recall figures are source-derived teaching points; use current local guidelines for clinical care.

Source material: F Pros pp. 16–17 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 3

Dental materials science

LO 1.9 | Question Bank Block C (Q161–290) | Block P (Q1161–1290) | Crash notebook Schematics 6–10 | Number card 1

IN ONE MINUTE

Every materials question is a property question: stiffness, dimensional stability, wetting or microstructure.

Co-Cr (220 GPa) is twice as stiff as gold, so its clasps engage 0.25 mm.

Glass ceramics: HF and silane. **Zirconia and base metals:** air abrasion and MDP.

PMMA: benzoyl peroxide initiates, hydroquinone inhibits; gaseous porosity sits in thick sections.

Materials were heavily examined last sitting: expect questions that give a clinical problem and ask which property or material explains it. Learn each material's composition, setting reaction, key property and one classic failure.

3.1 Mechanical and physical properties

TERM	MEANING	PROSTHODONTIC EXAMPLE
Elastic (Young's) modulus	Stiffness: slope of the elastic part of the stress–strain curve	Co-Cr (about 220 GPa) is twice as stiff as Type IV gold (about 90–100 GPa), so connectors can be thinner
Proportional / elastic limit	Highest stress with full recovery	Clasp must flex over the survey line without exceeding it
Yield strength	Stress at a set permanent strain (0.1–0.2%)	Resistance of a clasp to permanent distortion
Ultimate tensile strength	Maximum stress before fracture	—
Ductility	Permanent strain before fracture in tension	Low in Co-Cr: clasps cannot be adjusted repeatedly
Toughness	Energy absorbed to fracture (area under whole curve)	High-impact acrylic resists dropping
Resilience	Energy absorbed elastically (area under elastic part)	Soft liners, mouthguards
Fracture toughness (K1c)	Resistance to crack propagation	Zirconia much higher than glass ceramics
Fatigue	Fracture under repeated sub-maximal loads	Midline fracture of upper complete denture
Hardness (Vickers, Knoop, Brinell)	Resistance to indentation	Knoop for brittle materials and thin coatings
Creep / stress relaxation	Time-dependent strain under load / loss of stress	Waxes relax and distort if invested late
Viscoelasticity	Behaviour depends on rate of loading	Remove elastomeric impressions with a snap
Coefficient of thermal expansion	Dimensional change per °C	Tooth about 11 ppm/°C; acrylic about 80 ppm/°C; porcelain slightly lower than its metal substructure
Wettability (contact angle)	Low angle = hydrophilic	Polyether and modified PVS pour better without bubbles

Approximate elastic moduli: enamel 80–90 GPa; dentine 15–20 GPa; cortical bone 15–20 GPa; PMMA 2–3 GPa; composite 10–20 GPa; glass-fibre post 20–40 GPa; titanium 105–115 GPa; lithium disilicate about 95 GPa; zirconia about 200–210 GPa; Co-Cr about 220 GPa.

3.2 Impression materials

MATERIAL	TYPE	SETTING / KEY CHEMISTRY	KEY ADVANTAGES	KEY PROBLEMS
Impression plaster	Rigid, mucostatic	Hemihydrate → dihydrate	Records unloaded mucosa (flabby ridge)	Brittle; cannot record undercuts
Impression compound	Rigid, thermoplastic, mucocompressive	Physical (softens about 55–70°C)	Border moulding (greenstick), primary impressions	Low thermal conductivity: heat thoroughly; poor detail
Zinc oxide–eugenol paste	Rigid, mucostatic-ish	Chelation (zinc eugenolate)	Accurate secondary impressions of edentulous ridges	Undercuts; eugenol irritation; sticks to skin
Agar	Reversible hydrocolloid	Gel–sol by temperature	Hydrophilic, detailed	Needs conditioning bath and water-cooled trays; syneresis
Alginate	Irreversible hydrocolloid	Sodium alginate + calcium sulphate → calcium alginate; trisodium phosphate retards	Cheap, hydrophilic, quick	Syneresis and imbibition; low tear strength (needs at least 3–4 mm bulk); limited detail; pour within about 10–15 min unless extended-pour
Polysulphide	Elastomer, condensation	By-product water; lead dioxide catalyst	Long working time, high tear strength	Odour, mess, shrinkage over time: pour within an hour
Condensation silicone	Elastomer, condensation	By-product ethanol	Easy handling	Shrinkage as ethanol evaporates: pour promptly
Addition silicone (PVS)	Elastomer, addition	Vinyl–silane with platinum catalyst; no by-product	Best dimensional stability, 99.9% elastic recovery; can pour after days	Hydrophobic unless surfactant added; sulphur in latex gloves inhibits set; hydrogen release → wait about an hour before pouring (or palladium scavenger)
Polyether	Elastomer, cationic ring-opening	Imine ring opening; sulphonic ester initiator	Hydrophilic, accurate, good in moist field	Very stiff: hard to remove, may fracture stone dies; absorbs water (no prolonged immersion); bitter; occasional allergy

- **Pressure:** mucostatic (plaster, low-viscosity materials, spaced tray) records tissue at rest; mucocompressive (compound, close-fitting high-viscosity) records it under load. Selective pressure uses relief and windows.
- **Tray spacer (typical):** alginate about 3 mm; elastomers 1–3 mm depending on viscosity; ZOE close-fitting or about 0.5–1 mm.
- Always use the matching tray adhesive and a rigid tray: tray flexure gives distorted casts.
- **Disinfection:** rinse, then the material-specific disinfectant for the time on the label (usually hypochlorite- or aldehyde-free products); avoid prolonged immersion of alginate and polyether.
- **Putty–wash:** one-stage (simultaneous) risks putty displacing wash; two-stage risks distortion if the putty is reseated without adequate relief.

3.3 Gypsum products

ISO TYPE	PRODUCT	HEMIHYDRATE	WATER/POWDER RATIO (APPROX.)	USE
I	Impression plaster	β	0.50–0.75	Impressions
II	Model plaster	β (porous, irregular particles)	0.45–0.50	Mounting, study models
III	Dental stone	α (dense, regular, autoclaved)	0.28–0.30	Working casts for dentures
IV	Die stone, high strength, low expansion	α (modified)	0.22–0.24	Dies for crowns
V	Die stone, high strength, high expansion	α	0.18–0.22	Dies for base-metal alloys with high casting shrinkage

- **Reaction:** $CaSO_4 \cdot ½H_2O + 1½H_2O \rightarrow CaSO_4 \cdot 2H_2O$ (exothermic).
- More water → weaker, lower expansion, slower set. More spatulation → faster set, more expansion.
- **Accelerators:** potassium sulphate, gypsum seed (slurry), sodium chloride (low concentration). **Retarders:** borax, citrates, acetates; also blood and saliva (rinse impressions).
- Hygroscopic expansion occurs if the setting mass is immersed in water. Dry strength is about twice wet strength.

3.4 Waxes and investments

- **Inlay wax** has high thermal expansion and releases stresses (distortion) if left before investing: invest promptly.
- **Gypsum-bonded investment:** gold alloys only; calcium sulphate breaks down above about 700°C releasing sulphur dioxide.
- **Phosphate-bonded investment:** metal–ceramic and base-metal (Co-Cr, Ni-Cr) alloys; higher temperature.
- **Investment expansion** (setting, hygroscopic, thermal) compensates for alloy casting shrinkage (gold about 1.4%; Co-Cr about 2.3%).

3.5 Denture base acrylic (PMMA)

COMPONENT	POWDER	LIQUID
Main	PMMA beads (pre-polymerised)	Methyl methacrylate monomer
Initiator / inhibitor	Benzoyl peroxide (initiator)	Hydroquinone (inhibitor, shelf life)
Others	Pigments, fibres, opacifiers, plasticisers	Cross-linking agent (ethylene glycol dimethacrylate); tertiary amine activator in self-cure

- **Dough stages:** sandy → stringy (sticky) → dough (pack now) → rubbery → stiff.
- **Polymerisation shrinkage:** monomer alone about 21% by volume; dough about 7% volume, about 0.5% linear. Water uptake partly compensates.
- **Gaseous porosity:** heating too fast so monomer boils (100.3°C) → bubbles in the thickest section. **Contraction porosity:** too little dough or pressure → voids throughout and at the surface. **Granular porosity:** monomer evaporation from dough left uncovered.
- **Residual monomer:** heat-cured about 0.2–0.5%; self-cured (chemically activated, amine) about 2–5% — irritant and allergenic; self-cured is weaker and discolours (amine oxidation) but distorts less.
- **Glass transition temperature** about 105°C, but stress relief and warpage occur well below it: never clean dentures in hot water.
- **Fracture:** fatigue (midline fracture of upper dentures, stress concentration at the labial frenal notch, high palatal vault, flexion over a midline torus) or impact (dropping). Remedies: high-impact (butadiene–styrene rubber phase) acrylic, metal base,

glass-fibre reinforcement.

- **Pre-polymerised milled PMMA** (digital dentures): very low residual monomer and porosity, better fit, no polymerisation shrinkage.
- Acrylic is radiolucent and a poor thermal conductor (reduced perception of temperature).

3.6 Denture teeth, liners and cleansers

ITEM	KEY POINTS
Acrylic teeth	Chemical bond to base; easy to adjust; quiet; wear faster; preferred opposing natural teeth or gold
Porcelain teeth	Mechanical retention (pins anterior, diatoric holes posterior); wear resistant but abrade opposing teeth; click; brittle; need larger interarch space
Composite / IPN teeth	Improved wear resistance over acrylic, still bond chemically
Tissue conditioners	Short term (days to weeks); PEMA powder + aromatic ester plasticiser with ethanol; viscoelastic; for traumatised mucosa and functional impressions; harden as ethanol and plasticiser leach
Plasticised acrylic soft liner	Hardens over months as plasticiser leaches
Silicone soft liner	Longer lasting resilience; weaker bond to base; supports Candida; hard to trim and polish
Hypochlorite cleansers	Effective for acrylic; bleach and corrode Co-Cr and other metals
Alkaline peroxide cleansers	Safe for metal; less antifungal effect
Chlorhexidine	Antifungal soak; stains

3.7 Alloys

ALLOY	COMPOSITION AND ROLE OF ELEMENTS	KEY PROPERTIES	USES
Type IV gold (high noble)	Au, Ag, Cu (hardening by heat treatment), Pd, Pt	Modulus about 90–100 GPa; good ductility	Clasps (0.5 mm undercut), RPD frameworks (thicker than Co-Cr), crowns
Cobalt–chromium	Co about 60%; Cr about 25–30% (passive Cr_2O_3 film; needs at least 12–13%); Mo (grain refinement, strength); C (carbides → hardness; too much → brittle)	Modulus about 220 GPa; low density; low ductility; casting shrinkage about 2.3%	RPD frameworks; clasps need 0.25 mm undercut and at least about 15 mm length
Nickel–chromium	Ni, Cr, sometimes Be (toxic)	Similar to Co-Cr; nickel allergy (more common in women)	Metal–ceramic and RBB frameworks
Commercially pure titanium (Grades 1–4)	Ti with O, Fe; Grade 4 strongest CP grade	Modulus about 110 GPa; TiO_2 surface; biocompatible; difficult to cast	Implant bodies
Ti-6Al-4V (Grade 5)	Ti with aluminium, vanadium	Stronger than CP Ti	Abutments, screws, frameworks
Wrought stainless steel / wrought gold wire	Cold-worked, fibrous grain	Flexible, adjustable	Clasps engaging 0.5–0.75 mm undercut

- **Cast clasp undercut depths:** Co-Cr 0.25 mm; cast gold 0.5 mm; wrought wire 0.5–0.75 mm.
- **Corrosion:** galvanic (dissimilar metals in saliva, e.g., amalgam against gold), crevice, stress corrosion. Tarnish is surface discolouration only.

3.8 Metal–ceramic restorations

- **Bond mechanisms:** chemical (oxide layer: tin and indium oxides on noble alloys; chromium oxide on base metals) — the most important; mechanical (air-abrasion roughening); compressive (metal CTE slightly higher than porcelain so porcelain ends in compression on cooling); van der Waals.
- Too thick an oxide layer weakens the bond (Ni-Cr and Co-Cr form thick oxide).
- **Typical thickness:** metal 0.3–0.5 mm, opaque about 0.2 mm, body and enamel porcelain about 1 mm; labial reduction 1.2–1.5 mm.
- Porcelain should be supported by metal of uniform thickness and should not exceed about 2 mm unsupported (chipping).

3.9 All-ceramics

CLASS	EXAMPLE	FLEXURAL STRENGTH (APPROX.)	ETCHABLE WITH HF?	BONDING PROTOCOL
Feldspathic (glass)	Veneering porcelain, porcelain veneers	60–110 MPa	Yes	HF (about 5–10%), silane, resin cement (light-cure for veneers)
Leucite-reinforced glass ceramic	IPS Empress	120–160 MPa	Yes	HF, silane, resin cement
Lithium disilicate glass ceramic	IPS e.max	360–500 MPa	Yes (5% HF about 20 s)	Adhesive (HF + silane + resin) or conventional cement for crowns with good retention
Polycrystalline zirconia (3Y-TZP)	Monolithic or veneered	900–1200 MPa	No	Air-abrasion (50 µm alumina, low pressure) + MDP primer or MDP-containing cement; conventional (GIC/RMGIC) acceptable with retentive preparation
Translucent zirconia (4Y/5Y)	Anterior monolithic	600–800 MPa	No	As above; less transformation toughening
Resin-matrix ceramics	Hybrid ceramic, resin nano-ceramic	150–250 MPa	Varies	HF or air-abrasion + silane

- **Transformation toughening:** stress converts tetragonal to monoclinic zirconia with about 3–5% volume expansion, closing crack tips.
- **Low-temperature degradation (ageing):** spontaneous tetragonal → monoclinic change in moisture, roughening and weakening the surface.
- Zirconia is milled oversized in the pre-sintered (green or white) state, then sintered with about 20–25% linear shrinkage.
- Lithium disilicate is milled in the blue lithium metasilicate state, then crystallised.
- **Wear of opposing enamel:** unglazed or rough feldspathic porcelain is the most abrasive; polished monolithic zirconia is relatively kind. Always polish after adjustment.
- **Silane** is the coupling agent between silica-based ceramics and methacrylate resin; **MDP** bonds to metal oxides and zirconia.

3.10 Luting cements

CEMENT	RETENTION	KEY PROPERTIES	NOTES
Zinc phosphate	Mechanical only	Initial low pH (pulpal irritation);	Gold standard for retentive metal

		exothermic: mix incrementally on a cool slab	and metal–ceramic crowns
Zinc polycarboxylate	Chelation to calcium	Biocompatible; short working time; shear-thinning	Sensitive teeth
Glass ionomer	Ionic chemical bond	Fluoride release; CTE close to tooth; moisture sensitive early	Metal, metal–ceramic, zirconia
Resin-modified GIC	Ionic + micromechanical	Higher strength; hygroscopic expansion	Avoid under weak glass ceramics (expansion can fracture them)
Resin (etch-and-rinse, self-etch, self-adhesive)	Micromechanical + chemical (MDP, 4-META)	Highest bond strength; low solubility; technique sensitive	Essential for veneers, feldspathic and leucite ceramics, RBBs, inlays with poor retention
Zinc oxide–eugenol and non-eugenol temporary	Weak	Sedative; eugenol inhibits resin polymerisation	Use eugenol-free temporary cement before definitive resin bonding

ISO maximum film thickness for luting cements: **25 µm**.

3.11 Adhesion to tooth

- **Enamel:** 30–37% phosphoric acid for 15–30 s; three etch patterns (Type I prism cores, Type II prism peripheries, Type III mixed). Enamel bonding is the most reliable bond in dentistry.
- **Dentine:** smear layer; hybrid layer (Nakabayashi) of resin-infiltrated collagen; three-step etch-and-rinse is the reference standard; two-step self-etch with MDP performs well; universal adhesives can be used in either mode.
- Do not over-dry etched dentine (collagen collapse) or over-etch (deeper demineralisation than resin can infiltrate).
- **Sclerotic dentine** (common in cervical wear lesions) bonds less well: roughen and use selective enamel etch.

3.12 Resin-bonded bridge retainers (history and surface treatment)

- Rochette (perforated metal wing), Maryland (electrolytically etched metal), Virginia (lost-salt crystal), and today's air-abraded base-metal wings bonded with MDP-containing resin (e.g., Panavia).
- Noble alloys need tin plating or a metal primer to bond chemically.
- Silica coating (tribochemical) followed by silane is an alternative for metals and zirconia.

3.13 Posts

POST	MODULUS VS DENTINE	RETENTION	FAILURE MODE
Cast metal post and core	Much stiffer	Good with length; custom fit to non-circular canals	Root fracture (often unrestorable)
Prefabricated parallel-sided metal	Stiffer	More retentive and less wedging than tapered	Root fracture less likely than tapered
Tapered metal	Stiffer	Least retentive	Wedging → root fracture
Active (threaded)	Stiffer	Most retentive	High installation stress; avoid in thin roots
Glass/quartz fibre (bonded)	Similar to dentine	Adhesive bonding	Debonding (favourable, retrievable)

Guidelines: post length at least equal to crown height or two-thirds of root length; keep 4–5 mm of root filling apically; diameter no more than one-third of root width; at least 1 mm dentine around; ferrule 1.5–2 mm height of sound dentine circumferentially (most important factor for survival).

3.14 Implant and digital materials

- **Implant bodies:** CP titanium (Grade 4), titanium–zirconium alloy (narrow implants), zirconia (one-piece aesthetic option). Moderately rough surfaces (for example sand-blasted, acid-etched) speed osseointegration.
- **Abutments:** titanium (strongest, can show grey through thin tissue), zirconia (aesthetic, on titanium base), gold (cast-to).
- **CAD/CAM:** subtractive milling (zirconia, lithium disilicate, PMMA, Co-Cr, Ti) vs additive (stereolithography, digital light processing, selective laser melting for Co-Cr frameworks).
- Intraoral scanners are most accurate over short spans; full-arch and edentulous mucosa (mobile, featureless) scans are less reliable.

Examiner traps — Unit 3

1. Polyether is inherently hydrophilic; addition silicone is not, unless a surfactant is added.
2. Gypsum-bonded investment cannot be used above about 700°C.
3. Eugenol temporary cement inhibits later resin bonding.
4. A post retains the core; the ferrule, not the post, protects the root.

Previous
2 · Physiology and pathology

Next
4 · Pharmacology and the medically compromised

Choosing a restorative material

F Pros Chapter 11 → Diploma Unit 3

DIPLOMA LEARNING OUTCOMES 1.9 · 2c.6, 2c.13–2c.14

CORE PRINCIPLE

Most materials questions are clinical-choice problems: match properties, preparation, substrate and maintenance.

01 Bonding must match the substrate

Silica-based glass ceramics such as lithium disilicate require a suitable hydrofluoric-acid/silane-based protocol according to manufacturer directions. Zirconia is not etched in the same way; the F Pros source emphasises airborne-particle abrasion and MDP chemistry.

02 Preparations and restorability

A short, over-tapered crown preparation cannot be repaired by simply using more cement. Consider height, resistance form and an adhesive or auxiliary feature. A post retains a core; it does not strengthen a root-filled tooth.

03 What failures reveal

Layered ceramics are more vulnerable to chipping in some parafunctional settings; monolithic options may reduce the veneer fracture mode. Soft-tissue inflammation may indicate an overcontoured or deep subgingival margin rather than a defective ceramic itself.

04 Cost-benefit and consent

Compare repairability, retrievability, opposing-tooth wear, tooth reduction, aesthetics, laboratory workflow and replacement cost. Patient preference matters only after the clinical contraindications and maintenance requirements are explained.

05 Exam comparison

E-modulus tells you about stiffness, not toughness. Do not confuse flexural strength with wear behaviour, or high strength with a good bond. Treat the manufacturer-specific protocols as decisive.

SCOPE / GUIDANCE

Select treatment within competence. Do not convert laboratory handling principles into unsupervised specialist procedures.

Source material: F Pros pp. 33–34; 18–20 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 4

Pharmacology, pain control, emergencies and the medically compromised patient

LO 1.7, 2a.13, 2b.1–2b.2, 2b.11–2b.12, 2c.7–2c.8, 2c.15 | Question Bank Block D (Q291–360) | Block Q (Q1291–1360) | Crash notebook Schematics 23–24 | Number card 4

IN ONE MINUTE

Know the doses, never stop antiplatelets, and adjust DOACs only for higher-risk procedures.

Adrenaline 500 µg IM; **GTN** 400 µg; **aspirin** 300 mg; **glucagon** 1 mg IM.

Warfarin: treat if the INR is under 4, checked within 24 hours.

MRONJ: in higher-risk patients retain roots rather than extract; no routine drug holiday.

In SBAs on the medically compromised patient, the right answer usually keeps the drug going (antiplatelets, most anticoagulants, antiresorptives), uses local measures, and changes how the prosthesis is planned rather than refusing care. Doses below are adult and follow UK guidance (BNF, SDCEP, Resuscitation Council UK); always check the current versions.

4.1 Local anaesthetics

AGENT	PRESENTATION (UK CARTRIDGE)	POINTS TO REMEMBER
Lidocaine	2% with 1:80,000 adrenaline (2.2 mL = 44 mg)	Amide; reference agent; max dose with adrenaline commonly quoted 4.4 mg/kg in dental texts (BNF upper limit higher)
Articaine	4% with 1:100,000 or 1:200,000 adrenaline	Amide with an ester side chain: rapid plasma esterase metabolism, short half-life; good bone diffusion for infiltration; max about 7 mg/kg
Prilocaine	3% with felypressin 0.03 IU/mL	Use when adrenaline is a concern; felypressin has oxytocic action (avoid in late pregnancy); high doses cause methaemoglobinaemia
Mepivacaine	3% plain	Short duration; no vasoconstrictor

- **Toxicity:** CNS first (perioral tingling, tinnitus, drowsiness, convulsions), then cardiovascular depression.
- **Adrenaline interactions:** non-selective beta-blockers (hypertension with reflex bradycardia), tricyclic antidepressants, cocaine; limit dose in significant cardiac disease.
- **Failed inferior alveolar block causes:** accessory innervation (mylohyoid nerve, cervical plexus), bifid canal, infection (low tissue pH reduces non-ionised fraction), anxiety, poor technique. **Remedies:** Gow-Gates or Akinosi block, articaine buccal infiltration, intraligamentary.

4.2 Sedation (UK IACSD 2020 standards)

- **Inhalation sedation:** nitrous oxide in oxygen titrated; machines deliver at least 30% oxygen (maximum 70% N_2O); fail-safe and scavenging required. Relative contraindications: first trimester pregnancy, nasal obstruction, severe COPD, recent middle ear or eye surgery with gas, vitamin B12 deficiency.
- **IV sedation:** midazolam titrated; antagonist flumazenil. Escort for 24 hours (adults treated with IV), no driving or operating machinery until next day. Contraindications: pregnancy, severe COPD, severe liver disease, myasthenia gravis, allergy.
- **Monitoring:** pulse oximetry and blood pressure for IV; clinical monitoring for inhalation.
- Primary care is generally for ASA I–II patients; ASA III+ to secondary care.

4.3 Antimicrobials (SDCEP Drug Prescribing for Dentistry)

INDICATION	FIRST CHOICE	PENICILLIN ALLERGY
Spreading dental infection with systemic signs, after drainage attempted	Phenoxymethylpenicillin 500 mg four times daily for 5 days, or amoxicillin 500 mg three times daily for 5 days	Clarithromycin 500 mg twice daily (or metronidazole 400 mg three times daily for anaerobic infections)
Necrotising periodontal disease	Metronidazole 400 mg three times daily for 3 days	—
Denture stomatitis (if hygiene alone fails)	Miconazole oral gel; nystatin 100,000 units/mL; fluconazole 50 mg daily 7 days	—
Angular cheilitis	Miconazole cream (antifungal and anti-staphylococcal)	Sodium fusidate cream if staphylococcal

- Antibiotics are not a substitute for drainage or removal of the cause.
- **Miconazole** (including topical gel) and **fluconazole** potentiate warfarin (INR rise, bleeding) and increase statin levels (simvastatin myopathy).
- **Infective endocarditis:** NICE CG64 says antibiotic prophylaxis is not recommended routinely. SDCEP advises that for patients at special risk (previous endocarditis, prosthetic valves, certain congenital heart disease) the dentist should consider it in consultation with the cardiologist.
- **Chlorhexidine 0.2% mouthwash:** wait about 30 minutes after toothpaste (sodium lauryl sulphate inactivates it); causes extrinsic staining and taste disturbance.

4.4 Analgesia

- **Ibuprofen** 400 mg four times daily (with food). Avoid or use caution with peptic ulcer, renal impairment, anticoagulants, third-trimester pregnancy, NSAID-sensitive asthma.
- **Paracetamol** 1 g four times daily, maximum 4 g in 24 hours (reduce in low body weight, hepatic impairment, alcohol dependence).
- Combined ibuprofen and paracetamol is more effective than either alone.

4.5 Bleeding risk

DRUG	SDCEP APPROACH FOR DENTAL TREATMENT
Warfarin	Check INR, ideally within 24 hours (up to 72 hours if stable); treat in primary care if INR below 4; local haemostatic measures
Apixaban, dabigatran	Low-risk procedures: no interruption. Higher-risk procedures: miss the morning dose
Rivaroxaban, edoxaban	Higher-risk procedures: delay the morning dose until after treatment (4 hours after haemostasis)
Antiplatelets (aspirin, clopidogrel, ticagrelor), single or dual	Do not stop; local measures
Haemophilia / von Willebrand	Liaise with haemophilia centre; avoid IANB without factor cover

For prosthodontics: crown preparation with retraction cord, impressions and denture work are **low bleeding-risk** procedures.

4.6 Antiresorptive and antiangiogenic drugs (MRONJ)

- **Definition:** exposed bone, or bone that can be probed through a fistula, in the maxillofacial region for more than 8 weeks, in a patient with a history of antiresorptive or antiangiogenic therapy and no history of radiotherapy to the jaws or obvious metastatic disease.
- **Drugs:** bisphosphonates (alendronic acid, risedronate oral; zoledronic acid, ibandronic acid IV), denosumab (RANKL antibody — effect wears off in months, unlike bisphosphonates which persist in bone for years), antiangiogenics (bevacizumab, sunitinib).
- **Higher risk:** cancer patients on high-dose IV bisphosphonate or denosumab; osteoporosis patients on oral bisphosphonates over 5 years or with concurrent systemic glucocorticoids.
- **Prosthodontic implications:** get the mouth healthy before starting treatment; well-fitting, well-extended dentures with relief over bony prominences; avoid traumatic flanges; prefer root retention (decoronate, overdenture abutments, endodontics) to extraction; implants only after careful risk assessment and consent. SDCEP does not recommend drug holidays for routine dentistry.

4.7 Head and neck radiotherapy

- **Effects:** xerostomia (permanent if parotids in field), radiation caries (cervical, rampant), mucositis, trismus (fibrosis of muscles of mastication), candidosis, osteoradionecrosis (mandible far more than maxilla; risk lifelong, dose-dependent above about 50–60 Gy).
- **Before radiotherapy:** dental assessment; extract teeth of poor prognosis with time to heal (ideally 10–14 days or more before starting).
- **After:** 5000 ppm fluoride toothpaste, fluoride gel in custom trays, saliva substitutes, trismus exercises; dentures carefully adjusted with soft lining where mucosa is fragile; avoid extractions in the field if possible.
- Implants in irradiated bone have higher failure rates; these patients and maxillofacial (obturator) prostheses are Level 3.

4.8 Diabetes

- **Diagnosis:** HbA1c of 48 mmol/mol (6.5%) or above; prediabetes 42–47 mmol/mol.
- Two-way relationship with periodontitis; also candidosis, xerostomia, poor healing, burning mouth, higher implant complications when poorly controlled.
- The Level 2 curriculum expects the dentist to identify at-risk patients and refer to the GMP or diabetic services for HbA1c testing, per MCN policy.
- Book morning appointments after a normal meal; have oral glucose available.

4.9 Medical emergencies (Resuscitation Council UK)

EMERGENCY	KEY DRUG AND DOSE (ADULT)
Anaphylaxis	Adrenaline 1:1000 IM 500 micrograms (0.5 mL) into anterolateral thigh; repeat after 5 min if no improvement. Child 6–12 yrs 300 micrograms; under 6 yrs 150 micrograms
Angina / suspected myocardial infarction	GTN spray 400 micrograms sublingual; MI: aspirin 300 mg chewed, call 999, oxygen if hypoxic
Acute asthma	Salbutamol 100 micrograms per puff, 4–6 puffs via spacer, repeat (up to 10 puffs)
Hypoglycaemia	Conscious: 15–20 g oral glucose; unconscious: glucagon 1 mg IM
Prolonged seizure (5 minutes or more)	Buccal midazolam 10 mg
Vasovagal syncope	Lie flat, raise legs, oxygen
Adrenal crisis	Hydrocortisone (if prescribed and trained), call 999

Cardiac arrest	CPR 30:2 at 100–120 compressions/min, depth 5–6 cm; AED as soon as available
Choking (conscious, ineffective cough)	Up to 5 back blows then up to 5 abdominal thrusts

Oxygen: 15 L/min via non-rebreathe mask for the acutely ill.

4.10 Other conditions that change prosthodontic planning

CONDITION	PROSTHODONTIC IMPLICATIONS
Parkinson's disease, stroke, dementia	Poor neuromuscular adaptation: copy (replica) dentures, minimal changes; consider denture labelling (identification) in care homes
Epilepsy	Avoid prostheses that could be inhaled or fracture; metal-based or reinforced dentures; fixed options preferred; check drug-induced gingival overgrowth (phenytoin)
Xerostomia (drugs, Sjögren's)	Poor denture retention, soreness, root caries: high-fluoride, saliva substitute, consider implant retention
GORD, bulimia	Erosion: manage cause first; delay definitive restorations until stable; liaise with GMP
Pregnancy	Elective prosthodontics can usually wait; avoid prilocaine–felypressin; second trimester is the best time for necessary care; radiography only if justified
Renal failure / dialysis	Avoid NSAIDs and nephrotoxic drugs; treat on non-dialysis day; bleeding and infection risk
Immunosuppression, steroids	Candidosis, delayed healing; adrenal suppression rarely relevant for routine prosthodontics
Latex allergy	Non-latex gloves and dam; latex also inhibits addition silicone setting
Nickel or methacrylate allergy	Nickel-free alloys; for true MMA allergy consider fully cured heat-cured PMMA, polyamide or polycarbonate bases (patch test confirmation)
Cardiac pacemaker / ICD	Caution with electrosurgery and some ultrasonic devices
Osteoporosis	Not a contraindication to implants; antiresorptive drugs drive risk

Examiner traps — Unit 4

1. Prilocaine with felypressin: avoid in late pregnancy; large doses cause methaemoglobinaemia.
2. Propranolol with adrenaline gives hypertension with reflex bradycardia.
3. Infective endocarditis prophylaxis is not routine (NICE CG64).
4. Conscious hypoglycaemia: oral glucose. Unconscious: glucagon 1 mg IM.

Previous
3 · Dental materials science

Next
5 · Assessment, diagnosis and classifications

Medical risk and safe treatment sequencing

F Pros Chapters 1 and 4 → Diploma Unit 4

DIPLOMA LEARNING OUTCOMES 1.3, 1.7 · 2a.3, 2a.13–2a.14 · 2b.11–2b.12

CORE PRINCIPLE

The single best answer is often to recognise the modifier and adapt or defer care safely.

01 Assess the patient, not only the tooth

Record medical problems, allergies, anticoagulants, antiresorptives, immunosuppressants, glycaemic control, xerostomia and anxiety. Ask how each could affect treatment or healing rather than memorising a drug list.

02 Planning under uncertainty

When the patient has unstable disease or risk factors, stabilise oral disease, review medical information and decide whether liaison or referral is needed before elective definitive prosthodontics. Choose minimally invasive interventions when appropriate.

03 Communicate material risk

Include expected benefit, alternatives (including no treatment), maintenance costs, medical risk and any material clinical uncertainty in shared decision-making. Document the decision and any specialist advice.

04 Two traps

Uncontrolled periodontal disease is not a suitable foundation for new definitive bridgework. A patient on anticoagulants should not be instructed to stop medication simply because surgery is being considered; follow the relevant procedure-specific prescribing guidance.

SCOPE / GUIDANCE

Use the Diploma Unit 4 medication tables and up-to-date BNF/SDCEP/local emergency protocols for exact doses.

Source material: F Pros pp. 8–10, 16–17 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 5

Assessment, diagnosis, imaging and classifications

LO 2a.1–2a.21 | Question Bank Block E (Q361–440) | Block R (Q1361–1440) | Crash notebook Schematic 12 | Number cards 2 and 4

IN ONE MINUTE

Classify, stage and grade by the rules, then justify every radiograph.

Kennedy after planned extractions, with Applegate's rules.

BSP: stage by the worst site; grade by percentage bone loss divided by age.

IR(ME)R: the practitioner justifies, the operator exposes, the findings are recorded.

Classification questions are reliable marks: learn the exact definitions below, especially Kennedy with Applegate's rules, Cawood and Howell, BSP 2017 and the tooth wear indices.

5.1 History and examination essentials

- **History:** presenting complaint in the patient's words; expectations (aesthetic, functional); dental and denture history (age and number of previous dentures, what was good or bad about each); medical, drug, social (smoking, alcohol, diet, occupation, motivation, dexterity) and family history.
- **Extra-oral:** facial symmetry, lip support, lower face height, TMJ (clicks, crepitus, deviation, opening), muscles of mastication, lymph nodes, smile line (high smile line shows more than 3 mm of gingiva — aesthetic risk).
- **Intra-oral:** soft tissues and cancer screen; periodontal status (BPE, then full charting where codes 3 or 4); teeth and restorations; edentulous ridges (height, width, shape, flabby tissue, undercuts, tori); saliva; occlusion (RCP, ICP, slide, guidance, interferences, interocclusal space, OVD, freeway space).
- **Existing dentures:** retention, stability, support, extension, OVD, occlusion, aesthetics, hygiene, wear.
- **House's classification** of the edentulous patient's attitude: philosophical (best prognosis), exacting, indifferent, hysterical (sceptical).

5.2 Kennedy classification and Applegate's rules

CLASS	DESCRIPTION
I	Bilateral edentulous areas posterior to remaining natural teeth (bilateral free-end)
II	Unilateral edentulous area posterior to remaining natural teeth (unilateral free-end)
III	Unilateral bounded edentulous area with natural teeth in front and behind
IV	Single bilateral edentulous area anterior to the remaining teeth, crossing the midline

Applegate's rules

1. Classify after any planned extractions.
2. A missing third molar that will not be replaced is ignored.
3. A third molar present and used as an abutment is considered.
4. A missing second molar that will not be replaced is ignored.
5. The most posterior edentulous area always determines the class.
6. Other edentulous areas are modification spaces, designated by number.
7. The extent of a modification is not considered, only the number of additional areas.

8 Class IV has no modifications (any additional area behind it would make it a different class).

Craddock's support classification: Class 1 tooth-borne, Class 2 mucosa-borne, Class 3 tooth- and mucosa-borne.

5.3 Edentulous ridge classifications

GRADE	CAWOOD AND HOWELL (1988)	ATWOOD (1963)
I	Dentate	Pre-extraction
II	Immediately post-extraction	Post-extraction
III	Well-rounded ridge, adequate height and width	High, well-rounded
IV	Knife-edge ridge, adequate height, inadequate width	Knife-edge
V	Flat ridge, inadequate height and width	Low, well-rounded
VI	Depressed ridge with basal bone loss	Depressed

- **Palatal vault:** U-shaped (best retention and resistance to lateral displacement), V-shaped (poor stability, narrow post-dam), flat (poor resistance to lateral movement).
- **Soft palate (House):** Class I nearly horizontal (widest post-dam area, best), Class II about 45°, Class III steep about 70° (narrow post-dam area, poor posterior seal).
- **ACP Prosthodontic Diagnostic Index** classes I–IV grade complete and partial edentulism by ridge height, ridge relationship, muscle attachments and complicating factors.

5.4 Periodontal classification (BSP implementation of the 2017 classification)

ELEMENT	CRITERIA
Extent	Localised (under 30% of teeth), generalised (30% or more), molar–incisor pattern
Stage (worst site interproximal radiographic bone loss)	I: under 15% (or under 2 mm from CEJ); II: coronal third (15–33%); III: mid third; IV: apical third, usually with need for complex rehabilitation (fewer than 20 teeth, masticatory dysfunction, bite collapse)
Grade (% bone loss ÷ age)	A: under 0.5; B: 0.5–1.0; C: over 1.0
Current status	**Stable:** BOP under 10%, PPD 4 mm or less, no BOP at 4 mm sites. **In remission:** BOP 10% or more, PPD 4 mm or less, no BOP at 4 mm sites. **Unstable:** PPD 5 mm or more, or 4 mm with BOP
Risk factors	Smoking, diabetes control

- **BPE codes:** 0 healthy; 1 bleeding; 2 calculus or overhang; 3 pocket 3.5–5.5 mm (black band partly visible); 4 pocket over 5.5 mm; * furcation involvement.
- **Mobility (Miller):** Grade 1 slightly more than physiological; Grade 2 up to 1 mm horizontal; Grade 3 over 1 mm horizontal and/or vertical (depressible).
- **Furcation (Hamp):** I horizontal loss under 3 mm; II over 3 mm not through-and-through; III through-and-through.
- **NHS complexity** (Clinical standard for restorative dentistry 2022): after steps 1–2 in primary care, Stage III Grade A is Level 2; Stage III Grade B or C and all Stage IV are Level 3.

5.5 Tooth wear indices

SCORE	SMITH AND KNIGHT TWI (PER SURFACE)	BEWE (PER SEXTANT, WORST SURFACE)
0	No loss of enamel surface characteristics	No erosive wear
1	Loss of enamel surface characteristics	Initial loss of surface texture
2	Loss of enamel exposing dentine for under one-third of the surface (incisal: enamel loss only)	Distinct defect, hard tissue loss under 50% of surface
3	Loss of enamel exposing dentine for over one-third (incisal: dentine exposed)	Hard tissue loss of 50% or more of surface
4	Complete enamel loss, pulp or secondary dentine exposure	—

BEWE cumulative score (sum of six sextants)

SCORE	RISK	MANAGEMENT
2 or less	No or low	Routine maintenance
3–8	Low	—
9–13	Medium	Oral hygiene and dietary assessment, identify aetiology, monitor
14 or more	High	As medium, plus consider restoration

5.6 Occlusal classifications

- **Incisor relationship (British Standards Institute):** Class I — lower incisal edges occlude with or lie below the cingulum plateau of the upper incisors; Class II — lower incisal edges lie posterior to the cingulum plateau (division 1 proclined upper incisors with increased overjet; division 2 retroclined upper central incisors); Class III — lower incisal edges lie anterior to the cingulum plateau.
- **Angle molar classes I–III** (mesiobuccal cusp of upper first molar relative to buccal groove of lower first molar).
- **Skeletal pattern:** Class I, II, III by ANB angle (normal about 2–4°).
- **Crown–root ratio:** ideal 2:3; 1:1 is the minimum acceptable for an abutment.

5.7 Imaging (IRMER 2017 and FGDP selection criteria)

VIEW	BEST FOR	NOTES
Periapical (paralleling technique)	Abutment assessment: bone levels, crown–root ratio, periapical status, root form, root filling and post quality	Film holder and rectangular collimation reduce dose
Bitewing	Caries, restoration margins, crestal bone	Horizontal for caries; vertical for periodontal bone levels
Dental panoramic tomograph (OPG)	Overview of edentulous ridges, retained roots, unerupted teeth, pathology, ridge height, mental foramen position	Focal trough distortion; ghost images; not for fine detail
CBCT	Implant planning (bone height and width, inferior alveolar canal, sinus), complex root anatomy	Highest dose of dental options; justify individually; not for routine screening
Lateral cephalogram	Skeletal pattern, OVD planning in complex cases	Rarely needed at Level 2

- **IRMER roles:** referrer, practitioner (justifies), operator; employer sets procedures.
- Every exposure must be justified, optimised (ALARP) and clinically evaluated with a recorded report.

- **Effective doses (approximate):** intraoral 0.001–0.008 mSv; OPG 0.003–0.024 mSv; CBCT 0.02–1 mSv depending on field of view.

5.8 Special tests

- **Sensibility:** cold (tetrafluoroethane or ethyl chloride on a pledget; more reliable than heat), electric pulp test (assesses nerve response, not blood supply; false negatives after trauma, in immature, calcified or older teeth; false positives with anxiety or liquefactive necrosis). **True vitality** (blood flow): laser Doppler flowmetry, pulse oximetry.
- Percussion and palpation, mobility, bite tests (Tooth Slooth for cracked tooth), transillumination.
- **Mounted study casts** on a semi-adjustable articulator with facebow and an RCP record: assess interocclusal space, plan wax-up, survey for RPD design.
- **Diagnostic wax-up and mock-up:** test aesthetics, OVD, and communicate with the patient and laboratory.
- **Blood tests:** FBC (low MCV suggests iron deficiency; high MCV suggests B12 or folate deficiency), ferritin, B12, folate; HbA1c; INR for warfarin.

5.9 Prognosis and restorability

- Tooth prognosis depends on periodontal support, restorability (sound coronal dentine for a ferrule, caries extent), endodontic status, strategic value, and the patient's risk factors.
- **Restorability checklist:** at least 1.5–2 mm ferrule possible without crown lengthening that would compromise the root; no subgingival caries extending into the root beyond reach; crown–root ratio acceptable; no vertical root fracture; tooth is strategically useful.
- **Endodontic entry criteria in the NHS standard:** teeth must be restorable with sound coronal tissue above the crest (about 2 mm high and 1 mm wide, or over 30% of the original crown volume).

Examiner traps — Unit 5

1. A third molar used as an abutment counts in Kennedy classification.
2. Kennedy Class IV has no modifications.
3. A panoramic ghost image appears on the opposite side, higher and larger.
4. The electric pulp test measures nerve response, not blood flow; recently traumatised teeth may not respond.

Previous
4 · Pharmacology and the medically compromised

Next
6 · Removable partial dentures

Assessment and imaging: a decision pathway

F Pros Chapters 1–2 → Diploma Unit 5

DIPLOMA LEARNING OUTCOMES 2a.1–2a.21 · 2b.6

CORE PRINCIPLE

Select investigations only when their results may change the diagnosis, prognosis or plan.

01 History → examination → hypothesis

Start with the patient's concern, medical and dental history, risk factors and expectations. Examine soft tissue, hard tissue, periodontal support, denture-bearing tissues, tooth wear, function and occlusion. Generate a differential diagnosis before ordering further tests.

02 Choose the minimum useful imaging

Bitewings help with proximal caries and crestal bone; periapicals with periapical or root pathology; panoramic imaging for overview; CBCT only for an appropriate three-dimensional question that cannot be answered by lower-dose methods.

03 Interpret results together

A normal sensibility response does not prove vascular vitality, and radiographs have limitations. Periodontal attachment levels and clinical probing complement imaging. Read tests in the context of the history and other findings.

04 Record the implication

State whether a radiographic finding changes restorability, root configuration, bone support, anatomical safety or the feasible replacement options. Uncertain findings may require review, repeat tests or referral.

05 SBA traps

Do not select CBCT as routine screening; justify the field of view and radiation exposure. A normal radiograph does not overrule a concerning clinical examination or suspicious mucosal lesion.

SCOPE / GUIDANCE

The F Pros imaging chapter informs Diploma investigation decisions, not specialist-level CBCT reporting.

Source material: F Pros pp. 8–12 · Cross-checked against Diploma learning outcomes (Aug 2026)

Treatment strategies and consent

F Pros Chapter 3 → Diploma Unit 5

DIPLOMA LEARNING OUTCOMES 2b.3–2b.10 · 2c.2–2c.4, 2c.6, 2c.10

CORE PRINCIPLE

Stabilise disease first; choose a maintainable option that fits the patient's prognosis and priorities.

01 The five-phase plan

Urgent relief of pain or infection → stabilisation of periodontal/caries risk → re-evaluation → definitive restoration or replacement → supportive maintenance. A plan that starts with definitive crowns before disease control is usually unsound.

02 Compare options on common terms

For a missing unit, consider leaving the space, resin-bonded bridge, conventional bridge, removable prosthesis or implant-supported treatment when appropriate. Discuss tooth preparation, reversibility, aesthetics, cost, longevity, cleaning and failure consequences.

03 Conformative versus reorganised

Conform to a stable, comfortable existing intercuspal position where space and function permit. Reorganisation of occlusion, changes in OVD or complex multi-unit rehabilitation require careful diagnostic testing and often specialist involvement.

04 Consent is patient-specific

The supplied F Pros text references Montgomery: explain material risks and reasonable alternatives, including no treatment. The best choice is not necessarily the most technically elaborate option.

05 Communicate the sequence

Referral or laboratory instructions should contain the diagnosis, key-tooth prognosis, agreed phases, responsibilities, provisional plan, material/design choices and review arrangements.

SCOPE / GUIDANCE

NHS restorative standards distinguish moderate Level 2 planning from complex full-mouth/MDT Level 3 care.

Source material: F Pros pp. 13–15 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 6

Removable partial dentures

LO 2c.2–2c.6, 2c.13, 2c.20 | Question Bank Block F (Q441–550) | Block S (Q1441–1550) | Crash notebook Schematics 11–14 | Number card 2

IN ONE MINUTE

Saddles, support, retention, bracing, connection, review — and design every free-end saddle for movement.

Surveying defines one path of insertion; guide planes 2–4 mm parallel to it.

Rest seats: floor under 90°, 1–1.5 mm deep, cut after the guide planes.

Lingual bar needs 7–8 mm of functional sulcus; otherwise a plate.

The dentist is responsible for the design; the laboratory prescription must specify every component. A well-designed, tooth-supported cobalt–chromium RPD in a patient with good hygiene and regular recall does little harm to the remaining teeth (Bergman); mucosa-borne acrylic dentures with gingival coverage do.

6.1 Design sequence

1. **Saddles:** outline the edentulous areas to be restored (consider the shortened dental arch — not every space needs filling).
2. **Support:** tooth (rests), mucosa (saddle and palate coverage), or both.
3. **Retention:** direct (clasps, guide planes, attachments) and indirect.
4. **Bracing and reciprocation:** resist lateral forces; reciprocal arms and guide planes.
5. **Connection:** major and minor connectors.
6. **Review:** simplify, minimise gingival coverage, check aesthetics and hygiene.

The three forces to resist: displacement towards the tissues (support), away from the tissues (retention), and lateral (bracing).

6.2 Surveying

- **Surveyor parts:** vertical analysing rod, carbon marker (survey line), undercut gauges (0.25, 0.5, 0.75 mm), wax trimmer.
- **Survey line** = greatest convexity of a tooth relative to the chosen path of insertion. Undercut lies gingival to it.
- **Path of insertion** is chosen by: guide planes, retention (suitable undercuts), interference (avoid hard and soft tissue undercuts that block insertion), and aesthetics (minimise visible clasp metal, allow anterior teeth to fit without gaps).
- **Path of displacement** is assumed perpendicular to the occlusal plane; retention needs the path of insertion to differ from it (guide planes plus clasps).
- **Tilt:** lowering the front of the cast (heels up) tips the crowns mesially relative to the vertical rod and increases mesial undercuts; lowering the heels increases distal undercuts. Record the tilt by tripoding the cast so the laboratory can reproduce it.
- **Guide planes:** parallel proximal surfaces 2–4 mm high prepared in enamel (or built into crowns) parallel to the path of insertion; they define a single path, aid retention by friction, provide bracing and reduce food trapping.
- Height of contour may need modifying by enameloplasty (lower it) or composite (create an undercut or a rest seat).

6.3 Rests

- **Function:** support (transmit occlusal load along the long axis), maintain the vertical position of the denture, prevent gingival impingement, keep clasps in position, act as indirect retainers, prevent over-eruption.
- **Occlusal rest seat:** spoon-shaped with rounded floor inclined towards the centre of the tooth (angle with the minor connector

less than 90°); at least 1–1.5 mm deep for Co-Cr; about one-third of the buccolingual width; prepared after guide planes and before impressions.

- **Cingulum rests** on canines (natural or composite-built cingulum) are preferred to incisal rests (less visible, shorter lever, better load direction).
- Rests adjacent to free-end saddles are placed **mesially** on the abutment (RPI principle) so the abutment is tipped towards the arch rather than distally.
- Embrasure (double) rests and occlusal coverage rests may be used to restore the occlusal plane.

6.4 Clasps (direct retainers)

Requirements: retention, support (via its rest), reciprocation, encirclement (more than 180° of the tooth), passivity at rest, bracing.

CLASP	APPROACH	INDICATION	NOTES
Three-arm circumferential (Akers)	Occlusal	Bounded saddles, tooth-borne	Robust; more tooth coverage; retentive arm in lower third
Ring clasp	Occlusal	Mesially tilted lower molar with mesiolingual undercut	Needs supporting strut on the non-retentive side; long and flexible
Embrasure (double Akers) clasp	Occlusal	Side of Kennedy II with no modification space	Needs adequate embrasure space (prepare)
Reverse circlet / hairpin	Occlusal	Undercut next to saddle on the far side	Plaque trap; less used
I-bar, T-bar, Y-bar (Roach)	Gingival	Free-end saddles (RPI), aesthetic zones	Push-type retention, less tooth contact; contraindicated with shallow sulcus (under about 4 mm), soft tissue undercut, prominent frena
Wrought wire	Either	Undercuts of 0.5–0.75 mm; free-end saddle abutments (flexes, less torque)	Adjustable

- **Undercut engaged:** Co-Cr 0.25 mm, cast gold 0.5 mm, wrought wire 0.5–0.75 mm.
- **Flexibility** of a clasp increases with the cube of its length and decreases with the cube of its thickness: doubling length makes it about 8 times more flexible.
- **Reciprocal arm** sits on or above the survey line, opposite the retentive arm, and should contact the tooth before or with the retentive arm as it passes the survey line.
- **Retentive arm tip** in the gingival third, approached so it is passive when seated.

6.5 Free-end saddles (Kennedy I and II)

The problem — differential support. A tooth moves about 0.02–0.1 mm under load; mucosa compresses several times more (about 0.4–2 mm). The saddle rotates about the fulcrum line, torquing the abutment.

Solutions

- Maximum saddle extension: cover the retromolar pad and buccal shelf (mandible), tuberosity and hamular notch (maxilla).
- Mucodisplacive (functional) recording of the saddle area: altered cast technique (Applegate) or McLean's physiological impression.
- Mesial rests with an RPI or RPA clasp assembly, or a flexible wrought-wire clasp.
- Indirect retainers anterior to the fulcrum line.
- Narrow, reduced occlusal tables on the saddle teeth; harmonious occlusion.
- Stress-breakers (rarely used now).
- Regular review and relining as the ridge resorbs.

RPI (Kratochvil, Krol): mesial Rest, distal Proximal plate (guide plane), I-bar on the buccal surface at or mesial to the height of contour. When the saddle is loaded the I-bar and proximal plate move gingivally and mesially, disengaging the tooth and minimising torque.

RPA (Eliason): replaces the I-bar with an Akers-type circumferential arm when a gingivally approaching clasp is contraindicated.

6.6 Indirect retention

- **Fulcrum line:** line through the rests nearest to the free-end saddles (or through the principal abutments).
- A force pulling the saddle away from the ridge (sticky food) rotates the denture about this line; an indirect retainer placed as far anterior to the fulcrum line as possible resists it.
- **Examples:** rests on canines or premolars, cingulum rests, lingual plate, palatal plate extension anteriorly.
- Indirect retainers also give support for an anterior extension and prevent the major connector settling.

6.7 Major connectors

ARCH	CONNECTOR	INDICATION	KEY DIMENSIONS / NOTES
Maxilla	Palatal plate (full)	Long-span or bilateral free-end saddles, flat or resorbed palate, few remaining teeth	Maximum support and retention; most coverage
Maxilla	Mid-palatal strap	Short bounded saddles on both sides	At least 8 mm wide; thin and rigid
Maxilla	Anteroposterior strap (ring / closed horseshoe)	Large torus, anterior and posterior saddles	Rigid with less coverage; avoid torus
Maxilla	Horseshoe (U-shaped)	Anterior saddles only, large inoperable torus	Least rigid; avoid where possible
Mandible	Lingual bar	At least 7–8 mm from gingival margin to functional floor of mouth	Half-pear shape, thickest inferiorly; about 4 mm high and at least 3 mm clear of the gingival margins
Mandible	Lingual plate	Shallow lingual sulcus (under about 7 mm); mobile anterior teeth (splinting); future additions; indirect retention	Covers gingival margins: needs excellent hygiene; scalloped to sit on the cingula
Mandible	Sublingual bar	Shallow sulcus where a lingual bar would impinge	Lies deeper, in the sulcus; recorded in a functional impression
Mandible	Labial bar	Lingually inclined teeth, large mandibular tori	Rarely used; bulky lip
Mandible	Dental bar / continuous clasp (Kennedy bar)	Indirect retention, recession prevents lingual plate	Lies on cingula; food trap

- **Gingival margins:** either keep connector edges at least 3 mm (mandible) to 6 mm (maxilla) away, or cover fully with relief; never end a connector on the gingival margin.
- Major connectors must be rigid so forces spread across the arch.
- Minor connectors cross gingival margins at right angles, as narrow as possible, and lie in embrasures.

6.8 Acrylic vs cobalt–chromium

FEATURE	ACRYLIC RPD	CO-CR RPD
Support	Mainly mucosa (wire rests possible)	Tooth (cast rests) and mucosa

Bulk and coverage	Thick, covers gingival margins	Thin, rigid, open design
Periodontal and caries risk	Higher ("gum stripper")	Lower if well designed
Additions	Easy	Possible with retentive mesh; welding or soldering harder
Cost and time	Low, quick	Higher, more visits
Best use	Interim, immediate, transitional, teeth of uncertain prognosis	Definitive

- **Every denture (Every, 1949):** upper acrylic mucosa-borne design with point contact on teeth, open embrasures, free gingival margins, distal stabilising wires, posterior teeth in contact to prevent drift. Maximises gingival health in an acrylic RPD.
- **Spoon denture:** small upper anterior mucosa-borne denture with no gingival contact; an interim option.
- **Swing-lock denture:** hinged labial bar engaging labial undercuts of remaining teeth; for missing anterior teeth or periodontally compromised teeth where conventional clasping is poor.
- **Flexible (polyamide/nylon) dentures:** aesthetic, no metal clasps, but hard to adjust, reline or add to; mainly mucosa-borne.
- **Precision attachments:** intracoronal or extracoronal; good aesthetics; need crowned abutments and are usually beyond Level 2.

6.9 RPD occlusion, insertion and review

- Occlusion should be in harmony with the existing ICP; correct over-eruption, tilted teeth and occlusal plane before design (pre-prosthetic optimisation is Level 2).
- **Try the metal framework before setting teeth:** check fit with disclosing medium, rests seated, clasps passive, occlusion not raised.
- **Instructions:** insertion and removal along the path (not by clasp), cleaning (brush and soak; no hypochlorite on Co-Cr), night-time removal, review at about 1 week and then regular recalls.
- **Long-term:** monitor caries and periodontal health at abutments, reline free-end saddles as the ridge resorbs (check by pressing on indirect retainers — if the saddle lifts, reline).

Examiner traps — Unit 6

1. Free-end abutments take mesial rests.
2. Lowering the front of the cast increases mesial undercuts.
3. The horseshoe is the least rigid maxillary connector.
4. Hypochlorite corrodes Co-Cr; use alkaline peroxide.

Previous
5 · Assessment, diagnosis and classifications

Next
7 · Complete dentures

RPD design: applying the sequence

F Pros Chapter 6 → Diploma Unit 6

DIPLOMA LEARNING OUTCOMES 2c.2–2c.6, 2c.13, 2c.20

CORE PRINCIPLE

Start from the saddle and support; only then choose clasps, connector and indirect retention.

01 Design in order

Saddles → support → retention → bracing/reciprocation → major connectors → indirect retention → review. Identify Kennedy class after extractions; a distal-extension RPD needs tooth-and-mucosa support.

02 Biomechanical consequences

Mesial rests and stress-releasing concepts may reduce torque on distal-extension abutments. Survey casts, establish the insertion path and create suitable guide planes and rest seats before taking the definitive impression.

03 Numbers worth interpreting

The F Pros source gives a common 0.25 mm Co-Cr undercut and approximately 0.5 mm for cast gold/wrought wire. A lingual bar generally needs adequate functional sulcus depth, approximately 7–8 mm between the gingival margin and floor of mouth; otherwise consider a lingual plate.

04 Common failures

Pain after fitting may arise from overextension, tissue pressure or a high rest seat. Distal-extension saddles lose support as the ridge changes; assess the need for reline rather than merely tightening a clasp.

05 Exam trap

A retentive clasp without effective reciprocation is not a complete design. Choosing a clasp before assessing support and insertion path reverses the design logic.

SCOPE / GUIDANCE

Framework design must be individualized to teeth, anatomy and support; numerical rules are approximate.

Source material: F Pros pp. 21–23 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 7

Complete dentures

LO 2c.4, 2c.9, 2c.13, 2c.19, 2c.21–2c.27 | Question Bank Block G (Q551–650) | Block T (Q1551–1650) | Crash notebook Schematics 15–16 | Number card 2

IN ONE MINUTE

Retention is physics, stability is the neutral zone, and OVD is RVD minus about 3 mm.

Special trays 2 mm short, border moulded; post-dam along the vibrating line.

Record the horizontal relation in RCP; the Gothic arch apex marks it.

Every post-insertion complaint maps to a flange, the seal, the occlusion or the OVD.

Most complete denture SBAs are fault-finding: a symptom is given and you must name the clinical or technical cause. Learn the problem table in 7.9 thoroughly.

7.1 Retention, stability, support

- **Retention:** resistance to displacement away from the tissues. Physical factors (adhesion, cohesion, interfacial surface tension, viscosity, atmospheric pressure via the peripheral seal), muscular factors (shape of polished surfaces, neutral zone, tongue and cheek control), mechanical factors (undercuts, implants, adhesives).
- **Stability:** resistance to horizontal and rotational forces — ridge form, flange extension, occlusion (balanced, no deflective contacts), teeth in the neutral zone.
- **Support:** resistance to displacement towards the tissues — maximum extension over stress-bearing areas (Section 1.6).

7.2 Clinical stages

VISIT	STAGE	KEY POINTS
1	Primary impressions	Stock tray with alginate or impression compound; record full sulcus depth; prescribe special trays (spacer, extension 2 mm short of the sulcus to allow border moulding, handle not interfering with lips)
2	Secondary (master) impressions	Border mould with greenstick compound; ZOE, light-bodied silicone or polyether in the special tray; cast in Type III stone
3	Jaw registration	Record OVD, RCP, occlusal plane, centre line, canine lines, lip line (smile line), tooth mould and shade; facebow if semi-adjustable articulator
4	Try-in	Wax trial dentures: check extension, retention, OVD, RCP contacts, aesthetics, phonetics; patient approval; post-dam marked
5	Fit (insertion)	Pressure-indicating paste on fitting surface, check extension, occlusal refinement (clinical remount if needed), instructions
6	Review	About 1 week: adjust ulcers, occlusion

7.3 Record blocks and the occlusal plane

- **Upper rim:** labial face supports the lip; the labial surfaces of the central incisors usually lie about 8–10 mm anterior to the centre of the incisive papilla.
- **Upper incisal level:** about 1–2 mm of rim visible below the relaxed upper lip in younger patients; less with age.
- **Occlusal plane** parallel to the interpupillary line anteriorly and to the ala–tragal line (Camper's plane) posteriorly; checked with a Fox's plane guide.

- **Lower posterior occlusal plane** at about two-thirds the height of the retromolar pad and level with the lateral border of the tongue at rest.
- **Centre line** coincides with the facial midline (philtrum); canine lines at the corners of the mouth at rest; high lip (smile) line shows the necessary length of the teeth.

7.4 Vertical dimension

Rest vertical dimension (RVD) – occlusal vertical dimension (OVD) = freeway space (interocclusal rest space), normally 2–4 mm (average about 3 mm).

- **Measuring:** Willis gauge (base of nose to underside of chin; Willis noted pupil-to-stomion distance roughly equals nose-to-chin), dividers on dots, existing dentures, phonetics (closest speaking space about 1–2 mm on "s"), appearance, swallowing.
- **Rest position is not fixed:** affected by head posture, fatigue, emotion, dentures in or out — measure several times with the patient upright and relaxed.

OVD TOO HIGH (FREEWAY SPACE TOO SMALL)	OVD TOO LOW (FREEWAY SPACE TOO LARGE)
Generalised soreness of the denture-bearing area, especially the lower ridge	Angular cheilitis (folding at the commissures)
Teeth click during speech	Aged, "collapsed" appearance; prominent chin
Strained appearance, lips apart at rest	Reduced masticatory efficiency
Fatigue of masticatory muscles; difficulty eating	Cheek biting (cheeks fold between the teeth)
Faster ridge resorption	Tongue may seem crowded

7.5 Horizontal jaw relation

- Complete dentures are made in the **retruded contact position (RCP)** because it is reproducible; ICP is coincident with it in the finished dentures.
- **Recording:** guide the mandible gently to the retruded arc with the patient relaxed and the tongue curled back; check that the blocks meet evenly without sliding.
- **Gothic arch (needle-point) tracer:** the apex of the arrowhead marks RCP; useful when registration is unreliable.
- **Errors in RCP** show at try-in as posterior teeth not meeting evenly, or a slide. Take a new registration and re-articulate rather than adjusting teeth.

7.6 Articulators and occlusion

- **Simple hinge:** no lateral movement, ignore.
- **Average value:** fixed condylar guidance (about 30°) and Bonwill triangle (about 10 cm sides); adequate for most complete dentures.
- **Semi-adjustable (arcon or non-arcon):** condylar guidance set from a protrusive record; Bennett angle often from Hanau's formula L = H/8 + 12; used with a facebow.
- **Arcon:** condylar element on the lower member, anatomically correct (Whip Mix, Denar). **Non-arcon:** condylar element on the upper member (Dentatus, Hanau H2).
- **Facebow:** records the relationship of the maxillary arch to the hinge axis (arbitrary or kinematic), so arcs of closure on the articulator match the patient's.
- **Christensen's phenomenon:** separation of posterior teeth (or rims) in protrusion because the condyles move down the eminence. Bilateral balanced occlusion requires a compensating curve.
- **Hanau's quint:** condylar guidance, incisal guidance, cusp angle, plane of orientation, compensating curve.
- **Balanced (bilateral) occlusion:** simultaneous contacts on working and non-working sides in excursion — keeps complete dentures stable.

- **Lingualised occlusion:** only upper palatal cusps contact the lower central fossae; good stability with resorbed ridges and acceptable aesthetics.
- **Monoplane (zero-degree) teeth:** least horizontal force; for severely resorbed ridges, Class III relationships, poor neuromuscular control.

7.7 Tooth selection and setting

- **Anterior width:** canine lines and inter-alar width; central incisor width about one-sixteenth of bizygomatic width (Berry). Williams' face-form theory and Frush and Fisher's dentogenics (sex, personality, age) guide shape but are weakly evidence-based.
- **Shade:** darker and with characterisation for older patients; involve the patient; avoid very light "denture look".
- **Lower posterior teeth** over the ridge crest and in the neutral zone; never on the ascending slope of the ramus (the denture would be pushed forward).
- **Upper posterior teeth** may sit buccal to a resorbed maxillary ridge; set in crossbite where the maxilla is much narrower than the mandible.
- **Neutral zone:** area where tongue forces balance lip and cheek forces; record with tissue conditioner, impression compound or soft wax on a lower base during function (swallowing, speaking). Most useful for resorbed lower ridges and previously unstable lower dentures.
- Posterior horizontal overlap (buccal overjet) prevents cheek biting.

7.8 Post-dam and phonetics

- **Post-dam:** on the upper master cast, along the vibrating line from hamular notch to hamular notch, deepest (about 1–1.5 mm) either side of the midline, shallower over the median raphe. Functions: posterior seal, compensates for polymerisation contraction lifting the posterior border, reduces food entry and gagging.
- **Vibrating line** marked by asking the patient to say "ah"; the Valsalva manoeuvre identifies the anterior vibrating line.

SOUND	WHAT IT CHECKS
"F" and "v"	Upper incisal edges touch the wet–dry line of the lower lip — checks upper incisor length and labial position
"S"	Closest speaking space (about 1–2 mm) — checks OVD; whistling on "s" suggests narrow anterior palate or teeth too far palatal
"T" and "d"	Tongue on the palatal rugae area — excessive palatal thickness gives a lisp

7.9 Post-insertion problems and their causes

COMPLAINT	LIKELY CAUSES
Upper denture drops when talking or opening wide	Over-extended distobuccal flange (coronoid process), poor post-dam, over-extension at hamular notch, flange impinging on frena
Upper denture loose at rest	Inadequate post-dam or peripheral seal, under-extension, poor fit
Lower denture lifts	Over-extended lingual flange (mylohyoid) or retromolar region, teeth outside the neutral zone, tongue retracted
Generalised soreness	Excessive OVD, occlusal discrepancy (RCP error), parafunction, inadequate support, systemic factors (anaemia, diabetes)
Localised ulcer at sulcus	Over-extended flange or sharp edge
Ulcer on ridge crest	Occlusal interference, pressure spot, sharp underlying bone

Pain and numbness of lower lip	Denture pressing on the mental nerve (resorbed ridge)
Cheek biting	Lack of posterior buccal overjet, low OVD, upper and lower posterior teeth edge to edge
Clicking teeth	Excessive OVD, porcelain teeth, loose dentures
Gagging	Thick or over-extended (sometimes under-extended) posterior border, loose upper, anxiety; manage with training base, palateless implant designs, desensitisation
Food under denture	Poor peripheral seal, under-extension
Burning mouth	Denture stomatitis, xerostomia, haematinic deficiency, diabetes, very rarely monomer allergy
Angular cheilitis	Low OVD, Candida, haematinic deficiency
Upper midline fracture	Fatigue: flexure over a midline fulcrum (torus, poor fit), deep frenal notch, occlusal imbalance, opposing natural teeth

A **clinical remount** (new RCP record on the processed dentures, remounted on the articulator) corrects occlusal errors from processing more accurately than adjusting in the mouth.

7.10 Special complete denture techniques

- **Copy (replica) dentures:** duplicate the polished and occlusal surfaces of dentures the patient has adapted to, changing only what is necessary (fitting surface, OVD, occlusion). Best for older patients with poor adaptive capacity. The replica (wax teeth on shellac or acrylic base) acts as special tray and record block.
- **Immediate dentures:** flanged (better retention, stability and reline potential) vs open-face (socketed; better emergence appearance but poor retention and cannot be relined labially). Teeth are removed from the cast and the sockets lightly trimmed. Leave in for the first 24 hours; review next day; reline at about 3–6 months; replace later as resorption continues.
- **Overdentures:** retained roots (canines are ideal — longest roots, strategic position) reduced to 2–3 mm above the gingiva with domed surfaces and root fillings; preserve bone and proprioception, improve support, retention with attachments. Risks: caries and periodontitis — fluoride gel in the denture fitting surface, meticulous hygiene.
- **Implant overdentures:** McGill consensus (2002) and York consensus (2009): a two-implant mandibular overdenture should be the minimum standard of care for the edentulous mandible.
- **Single complete denture against natural teeth:** risk of fracture and of combination syndrome (Kelly): anterior maxillary resorption and flabby ridge, tuberosity enlargement, papillary hyperplasia, over-eruption of lower anteriors, resorption under lower free-end saddles. Prevent by restoring posterior support (lower RPD) and reducing anterior loading.
- **Flabby ridge impressions:** mucostatic (impression plaster) or selective pressure (windowed special tray: mucocompressive material for the firm ridge, then low-viscosity material or plaster through the window over the flabby tissue).
- **Reline** (new fitting surface) vs **rebase** (whole base replaced). Chairside hard reline is quick but more porous; closed-mouth laboratory reline can increase OVD if the patient bites too firmly.
- **Pre-prosthetic surgery** (tuberosity reduction, vestibuloplasty, torus removal) is usually Level 3.

Examiner traps — Unit 7

1. Clicking teeth and generalised soreness mean the OVD is too high.
2. An upper denture that drops on wide opening points to the distobuccal flange and the coronoid.
3. The occlusal plane is parallel to the interpupillary line anteriorly and to Camper's plane posteriorly.
4. The two-implant overdenture is the minimum standard for the edentulous mandible.

Previous
6 · Removable partial dentures

Next
8 · Fixed prosthodontics

Complete dentures: stability and adaptation

F Pros Chapter 6 → Diploma Unit 7

DIPLOMA LEARNING OUTCOMES 2c.4, 2c.9, 2c.13, 2c.19, 2c.21–2c.27

CORE PRINCIPLE

Retention is only one part of success: harmony with the neutral zone and the patient's adaptation is equally important.

01 Clinical sequence

Primary impressions → special trays → border-moulded secondary impressions → occlusal rims and jaw registration → trial dentures → insertion → review. Recheck each step instead of hoping the technician will compensate for a faulty record.

02 Choose an impression strategy

Displace flabby tissue as little as clinically appropriate; selective pressure or a window technique may help. Functional borders and the posterior palatal seal have to be assessed clinically.

03 Occlusal scheme and OVD

The source summarises OVD as rest vertical dimension minus roughly 2–4 mm of freeway space. In severely resorbed ridges, lingualised or carefully balanced occlusion and a neutral-zone approach may improve stability.

04 Patient adaptation

For older patients with successful old dentures, copy techniques preserve useful adaptation while changing only identified defects. Explain the likely need for early relines with immediate dentures and the adjustment period following insertion.

05 Exam trap

Increasing vertical dimension or adding more adhesive rarely cures an overextended flange, inaccurate border, unstable record or incorrect neutral-zone position.

SCOPE / GUIDANCE

Discuss implant-assisted dentures as an option or referral; specialist implant surgery is not a Level 2 learning requirement.

Source material: F Pros pp. 21–23 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 8

Fixed prosthodontics

LO 2c.2, 2c.4, 2c.6, 2c.11, 2c.13–2c.14, 2c.20 | Question Bank Block H (Q651–760) | Block U (Q1651–1760) | Crash notebook Schematics 17–19 | Number card 3

IN ONE MINUTE

Taper, height and ferrule hold a crown on; Ante's law and the cube of the span decide a bridge.

Margins supragingival where possible, and at least 3 mm from the bone crest.

For sound abutments with good enamel, a resin-bonded cantilever is the first choice.

In ceramic connectors, height matters more than width.

Fixed bridge design was examined last sitting: be fluent in abutment evaluation, choosing between fixed-fixed, cantilever, fixed-movable and resin-bonded designs, pontic forms and the mechanics of span and connectors.

8.1 Principles of tooth preparation (Shillingburg)

1. Preservation of tooth structure (and pulp).
2. Retention and resistance form.
3. Structural durability of the restoration (adequate material thickness, functional cusp bevel).
4. Marginal integrity.
5. Preservation of the periodontium.

- **Taper (total occlusal convergence):** theoretical ideal about 6°; clinically 10–20° is typical and acceptable. Retention falls sharply as convergence increases.
- **Minimum axial wall height:** about 3 mm for incisors and premolars, 4 mm for molars. Height-to-width ratio of at least 0.4 improves resistance.
- **Short or over-tapered preparations:** add grooves or boxes, increase surface area, use adhesive cements, consider crown lengthening (Level 3 surgery) or elective endodontics with a core only as a last resort.
- **Pulp protection:** sharp burs, copious water spray, light intermittent pressure, avoid desiccation, well-fitting temporaries; dentine sealing.

8.2 Reduction guide

RESTORATION	AXIAL	OCCLUSAL / INCISAL	MARGIN
Full gold / cast metal	0.5–1 mm	1.5 mm functional cusp, 1 mm non-functional	Light chamfer (0.5 mm)
Metal–ceramic	Labial 1.2–1.5 mm; lingual 0.5–0.7 mm	1.5–2 mm	Labial shoulder (with or without bevel) or heavy chamfer; lingual chamfer
Lithium disilicate crown	1–1.5 mm	1.5–2 mm	Rounded shoulder or deep chamfer about 1 mm; rounded internal angles
Monolithic zirconia crown	0.5–1 mm	About 1 mm	Light chamfer or feather edge possible
Porcelain laminate veneer	0.3–0.7 mm labial (stay in enamel)	1–2 mm incisal reduction if overlapped	Chamfer at or just above gingival margin
Onlay	As required	1.5–2 mm cuspal	Butt or chamfer

Functional (supporting) cusps: palatal of upper and buccal of lower posteriors; give them a functional cusp bevel.

8.3 Margins and the periodontium

- **Prefer supragingival margins** (hygiene, impression, finishing, inspection). Equigingival or slightly subgingival only for aesthetics, caries, existing restorations, ferrule or retention.
- **Supracrestal tissue attachment ("biologic width")** averages about 2 mm (connective tissue about 1 mm plus junctional epithelium about 1 mm; Gargiulo); with the sulcus, keep margins about 3 mm from the bone crest. Invasion causes chronic inflammation, recession or bone loss.
- Overcontoured crowns (especially in furcation areas) trap plaque: keep emergence profiles flat.
- **Gingival retraction:** plain or impregnated cord (aluminium chloride or ferric sulphate; avoid adrenaline cord in cardiovascular disease), double-cord technique, retraction pastes, electrosurgery or laser.

8.4 Temporary (provisional) restorations

- **Purposes:** protect the pulp, maintain tooth position (prevent drift and over-eruption), protect gingival health, restore function and aesthetics, test the planned shape, OVD or occlusion.
- **Materials:** bis-acryl composite (low exotherm, good strength, common); PMMA (stronger for long-term but exothermic and higher residual monomer); prefabricated polycarbonate or metal crowns.
- Use eugenol-free temporary cement if the definitive restoration will be resin-bonded.

8.5 Occlusion, shade and impressions

- **Conformative approach** (most single units and short bridges): restore to the existing ICP without changing the occlusal scheme.
- **Reorganised approach** (RCP/centric relation): when ICP is unstable, extensive restoration is planned, or OVD will change. Usually Level 3 when major.
- **Shade:** take before preparation (teeth dehydrate and lighten during treatment), in colour-corrected daylight, quickly; value is the most important dimension; photograph with shade tabs and send to the laboratory.
- **Impressions:** addition silicone or polyether in a rigid tray; dual-arch (triple) trays only for single units with stable intercuspation. Intraoral scans suit short-span work.
- **Die spacer** (about 20–40 µm) gives room for cement; it stops short of the margin.

8.6 Endodontically treated teeth

- **Posterior root-filled teeth need cuspal coverage** (onlay or crown) — survival improves markedly (Sorensen and Martinoff); anterior root-filled teeth with little tissue loss can be restored with composite.
- **Ferrule** of 1.5–2 mm of sound dentine is the most important factor in fracture resistance (Section 3.13 for post rules).
- Posts retain cores; they do not strengthen roots.
- **Molars:** amalgam or composite core retained in the pulp chamber (Nayyar core) without a post is often enough.
- Provide a good coronal seal promptly after root canal treatment; coronal leakage contributes to failure.

8.7 Bridge terminology and abutment evaluation

- **Terms:** abutment (supporting tooth), retainer (restoration on the abutment), pontic (replacement tooth), connector (joint), span, unit.
- **Abutment assessment:** periodontal health and bone support, crown–root ratio (ideally 2:3, minimum 1:1), root form (divergent multi-rooted and long roots are better than conical or fused), root surface area, mobility, pulpal and endodontic status,

restorative condition, position and angulation, opposing occlusion.

- **Ante's law:** the total root surface area of the abutments should equal or exceed that of the teeth being replaced. It is historical and not well supported by evidence, but still examined.
- **Relative root surface area (largest to smallest):** maxilla 6 > 7 > 3 > 4 ≈ 5 > 1 > 2; mandible 6 > 7 > 3 > 5 ≈ 4 > 2 > 1. Upper canine has the largest root of the single-rooted teeth.
- **Span mechanics:** deflection of a beam increases with the cube of its length and decreases with the cube of its thickness (occluso-gingival connector height). Doubling the span gives about eight times the flexure.
- **Arch curvature:** pontics lying outside the line joining the abutments (e.g., replacing all four upper incisors) act as a lever arm causing torque.

8.8 Bridge designs

DESIGN	DESCRIPTION	INDICATIONS	PROBLEMS
Conventional fixed-fixed	Rigid connectors at both ends; preparations must share one path of insertion	Bounded spans with sound abutments needing crowns anyway	Most tooth destruction; parallel preparations difficult with tilted teeth; loss of retention of one retainer goes unnoticed (caries)
Conventional cantilever	One abutment (or splinted abutments) on one side only	Short spans; e.g., upper canine supporting a lateral incisor pontic; central supporting a lateral	Torque on the abutment; pontic should be no larger than the abutment; avoid long or posterior distal cantilevers with heavy occlusion
Fixed-movable (non-rigid)	Rigid connector to the major retainer; key-and-keyway (dovetail) movable joint at the minor retainer	Abutments not parallel (tilted molar); pier abutments; long spans	Keyway placed in the distal surface of the mesial (minor) retainer so that mesial drift seats the key; needs adequate retainer bulk
Spring cantilever	Palatal bar from a posterior abutment to a distant anterior pontic	Historic: replacing an upper incisor with diastema	Palatal bar unhygienic; rarely used
Resin-bonded (adhesive), cantilever	Single metal or zirconia wing bonded to enamel of one abutment	First choice for single missing incisors and premolars with sound abutments; young patients	Debond (rebond if intact); needs enamel and space; less effective where bruxism or heavy occlusion
Resin-bonded, fixed-fixed	Wings on both abutments	Rarely preferred now	Differential movement → one wing debonds unnoticed → caries; lower survival than cantilever RBB
Hybrid	Conventional retainer at one end, resin-bonded at the other, usually with a movable joint	Selected cases	Complex

Pier abutment: a rigid five-unit bridge with a pier tooth risks debonding of a terminal retainer because the pier acts as a fulcrum. Use a non-rigid connector (keyway on the distal of the pier retainer) or separate bridges.

8.9 Resin-bonded bridges in detail

- **Advantages:** minimal or no preparation, no anaesthesia, reversible, cheap, quick; pulp not at risk; failures usually rebondable.
- **Requirements:** sound abutment with large enamel area, adequate occlusal clearance for the wing (about 0.7 mm Co-Cr or Ni-Cr) or a Dahl approach, stable occlusion.
- **Design features:** maximum wing coverage of enamel within aesthetic limits, wraparound (about 180°), occlusal or cingulum rests, minimal preparation (rest seats, guide planes) to define the path and resistance.
- **Bonding:** air-abrade the metal (50 µm alumina), MDP-containing resin cement (e.g., Panavia) for base metals; rubber dam; etch enamel.

- **Survival is good:** single-retainer cantilever designs show at least as good survival as fixed-fixed and fail more safely.

8.10 Pontic design

PONTIC	RIDGE CONTACT	USE	NOTES
Hygienic (sanitary)	None (at least 3 mm gap)	Lower posterior, non-aesthetic zone	Easiest to clean
Conical (bullet, ovoid tip)	Point contact at the ridge crest	Lower posterior with thin ridges	Cleanable
Modified ridge lap	Labial (buccal) side only, T-shaped contact	Upper anterior and premolar aesthetic zones	Best balance of aesthetics and cleansability
Ovate	Sits in a concavity in the ridge	Highest aesthetic demand; after socket preservation or ridge shaping	Needs adequate ridge thickness; meticulous hygiene
Saddle / ridge lap	Concave, covers the ridge	Avoid	Unhygienic: inflammation and food trapping

- **Pontic tissue surface:** convex, smooth and glazed porcelain or highly polished metal; never pressing on the mucosa (blanching) in fixed work.
- Narrow the buccolingual width of posterior pontics to reduce load.

8.11 Connectors and all-ceramic bridges

- **Connectors are the weakest point** of all-ceramic bridges. Approximate minimum cross-sections: lithium disilicate 12 mm^2 anterior and 16 mm^2 premolar (3-unit maximum, not beyond the second premolar); zirconia about 9 mm^2 for 3-unit posterior and more for longer spans.
- **Height matters more than width:** connector strength rises with the square of height; deflection falls with the cube.
- Open gingival embrasures in connectors aid hygiene; rounded connector radii reduce stress concentration.

8.12 Veneers, onlays and inlays

- **Porcelain veneer indications:** discolouration not responsive to bleaching, minor malalignment, diastema closure, shape change, minor fractures.
- **Contraindications:** inadequate enamel, bruxism or heavy occlusion, edge-to-edge or Class III incisor relationship, poor oral hygiene, heavily restored teeth.
- **Designs:** window (no incisal involvement), feathered incisal, butt joint, incisal overlap (palatal finish line — most common for stress distribution and seating).
- **Bond** with HF-etched, silanated ceramic and light-cured resin cement on etched enamel.
- **Onlays:** cusp coverage for teeth with weakened cusps (cusp width less than about half its height); adhesive onlays preserve more tooth than crowns.
- **Inlay cavities** diverge occlusally (about 6–10°) for a path of withdrawal; no undercuts.

8.13 Laboratory communication and regulation

- **The prescription is a legal document:** patient identifier, clinician details, materials, design, shade, margin, pontic form, connector type, delivery date.
- Custom-made devices fall under the UK Medical Devices Regulations: the laboratory provides a statement of manufacture; the manufacturer must be registered with the MHRA. Retain records.
- Dental technicians and clinical dental technicians are GDC registrants with their own scope of practice.

Examiner traps — Unit 8

1. A pier abutment needs a non-rigid connector, not a stiffer bridge.
2. In a fixed-movable bridge the keyway sits in the distal of the mesial (minor) retainer.
3. The saddle (ridge lap) pontic is unhygienic; avoid it.
4. Take the shade before preparation; value matters most.

Previous
7 · Complete dentures

Next
9 · Occlusion and TMD

Fixed prosthodontics: preparation to review

F Pros Chapter 5 → Diploma Unit 8

DIPLOMA LEARNING OUTCOMES 2c.2, 2c.4, 2c.6, 2c.11, 2c.13–2c.14

CORE PRINCIPLE

Preserve sound tooth structure and select a restoration with adequate retention, resistance and long-term hygiene.

01 Preparation principles

Compare partial coverage, onlay/overlay, resin-bonded bridge and full crown. Verify clearance, marginal design, axial wall height, taper and ferrule rather than treating one preparation number as a universal rule.

02 Use the right adhesive chemistry

The F Pros notes contrast silica-based ceramics requiring silane-based bonding with zirconia requiring alternative surface conditioning/MDP approaches. A cement cannot compensate for poor resistance form.

03 Biological protection

Deep reduction, heat, desiccation and provisional leakage can damage the pulp. Subgingival overcontours and poor emergence profiles increase plaque retention and soft-tissue inflammation.

04 Bridge design choices

Review span, abutment periodontal support, tooth preservation, patient cleaning ability and the feasibility of a cantilever resin-bonded design. More retainers are not automatically safer.

05 Success is more than survival

At recall inspect biological health, contacts, margins, caries, cracks, occlusion, aesthetics and patient satisfaction. Parafunction may change material and maintenance choices.

SCOPE / GUIDANCE

Full-mouth reconstruction and complex occlusal reorganisation are Level 3/referral scenarios, not routine Diploma execution.

Source material: F Pros pp. 18–20 · Cross-checked against Diploma learning outcomes (Aug 2026)

Aesthetic dentistry: least invasive first

F Pros Chapter 10 → Diploma Unit 8

DIPLOMA LEARNING OUTCOMES 2a.16 · 2b.6, 2b.10 · 2c.4

CORE PRINCIPLE

Start with a systematic assessment of the smile, then solve the patient's actual concern using the least destructive option.

01 Analyse from outside inward

Facial symmetry and midline → lip support/smile line → incisal display and occlusal plane → tooth proportions, shade, gingival contour and papillae. A mathematical "golden proportion" is not a mandatory natural smile rule.

02 Treatment ladder

For colour: investigate aetiology and consider whitening first. For white spots: consider infiltration or microabrasion. For spacing or peg laterals: consider additive composite; for malalignment: orthodontics before elective tooth preparation.

03 Before an indirect veneer

Use diagnostic wax-up/mock-up, assess enamel availability, parafunction and deep overbite, and discuss irreversibility and long-term replacement. Whitening may alter the final shade and immediate resin bonding conditions.

04 Patient-centred consent

Discuss maintenance, failure, costs and reasonable alternatives including no treatment. Distinguish a realistic restorative concern from expectations that cannot safely be met.

05 SBA trap

A crown on an otherwise healthy tooth for a minor aesthetic complaint is generally less conservative than whitening, orthodontics or additive composite if those can meet the objective.

SCOPE / GUIDANCE

Complex gingival/orthognathic procedures require appropriate specialist assessment, not Diploma-level operative detail.

Source material: F Pros pp. 31–32 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 9

Occlusion and temporomandibular disorders

LO 2a.15, 2a.20, 2c.19 | Question Bank Block I (Q761–820) | Block V (Q1761–1820) | Crash notebook Schematics 20–21 | Number card 3

IN ONE MINUTE

Most TMD is myalgia: explain, self-care, simple analgesia and a hard full-arch splint, after excluding red flags.

Closed lock (DDwoR) deflects the mandible to the affected side.

Canine guidance discludes all other teeth; **group function** shares the load on the working side.

Bruxists: hard night splint, and monolithic zirconia or metal restorations.

For TMD, the evidence-based answer is almost always reversible, conservative care first (education, self-care, a stabilisation splint); irreversible occlusal adjustment is not a treatment for TMD.

9.1 Definitions

TERM	DEFINITION
Intercuspal position (ICP)	Maximum intercuspation, independent of condylar position
Retruded contact position (RCP)	First tooth contact when the mandible closes on the retruded (terminal hinge) arc
Centric relation (CR)	Condyles in their most superior-anterior position against the posterior slopes of the articular eminences, discs correctly interposed; a jaw-to-jaw relation independent of teeth
Slide	Movement from RCP to ICP; present in about 90% of people, usually about 1 mm forwards and upwards
Rest position	Mandible at rest with the teeth apart (freeway space 2–4 mm)
Anterior guidance	Contact of anterior teeth that guides the mandible in protrusion and lateral excursion
Canine guidance	Only canines contact on the working side in lateral excursion
Group function	Several teeth (canine, premolars, sometimes molars) share working-side contact
Mutually protected occlusion	Anterior teeth protect posteriors in excursions; posteriors protect anteriors in ICP
Working-side interference	Posterior contact on the working side that prevents canine or group guidance
Non-working (balancing) side interference	Contact on the side the mandible moves away from; most damaging in natural dentitions; desirable only in complete dentures (balanced occlusion)
Deflective contact	Contact that diverts the mandible from its normal path of closure
Bennett movement / angle	Lateral bodily shift of the mandible in excursion / angle of the non-working condyle path to the sagittal plane (about 7–15°)
Curve of Spee / curve of Wilson	Anteroposterior / mediolateral curvature of the occlusal plane

- **Posselt's envelope of motion:** maximum opening about 40–55 mm (under 40 mm is restricted), lateral excursions and protrusion about 8–10 mm each; pure hinge rotation gives about 20–25 mm incisal opening.
- **Features of an "ideal" (functional) occlusion:** stable ICP with even simultaneous contacts, axial loading of posterior teeth, anterior guidance that discludes posterior teeth, no non-working-side interferences, smooth slide or coincident RCP and ICP.

9.2 Diagnostic Criteria for TMD (DC/TMD, 2014)

GROUP	DIAGNOSIS	KEY FEATURES
Pain	Myalgia (local, myofascial, myofascial with referral)	Pain in masticatory muscles, familiar pain on palpation or opening; commonest TMD
Pain	Arthralgia	Joint pain on palpation or movement
Pain	Headache attributed to TMD	Temporal headache modified by jaw function
Joint	Disc displacement with reduction	Reciprocal click on opening and closing; disc recaptured
Joint	DDwR with intermittent locking	Click plus episodes of limited opening
Joint	Disc displacement without reduction, with limited opening (closed lock)	Opening under 40 mm; deflection to the affected side on opening; restricted contralateral excursion
Joint	DDwoR without limited opening	History of lock, now opening at least 40 mm
Joint	Degenerative joint disease	Crepitus; radiographic changes
Joint	Subluxation / dislocation	Open lock: condyle anterior to the eminence, unable to close

- **Aetiology** is multifactorial (biopsychosocial): parafunction, trauma, stress, anxiety and depression, sleep disturbance, widespread pain conditions. Occlusal factors have a weak association.
- TMD is commonest in women aged about 20–40.
- A click without pain or locking needs no treatment beyond reassurance.

9.3 Management

1. **Explanation and reassurance;** self-limiting in most patients.
2. **Self-care:** soft diet, avoid wide opening and gum chewing, warm compresses, jaw rest, stop habits (nail biting, clenching — "lips together, teeth apart").
3. **Simple analgesia** (NSAIDs short term).
4. **Stabilisation splint:** full-coverage hard acrylic (often maxillary, Michigan type), even contacts in RCP, canine guidance, worn at night.
5. **Physiotherapy,** jaw exercises, CBT and stress management.
6. **Referral (secondary care):** persistent pain, low-dose tricyclic (by GMP or specialist), botulinum toxin, arthrocentesis, arthroscopy, surgery.

- **Soft splints** are for short-term use (may increase clenching in some).
- **Anterior repositioning splints** (for disc displacement with reduction) are short-term only: long-term use risks a posterior open bite.
- **Partial-coverage anterior bite planes** can let uncovered posterior teeth over-erupt if worn continuously.
- **Closed lock of recent onset:** manipulation to recapture the disc may be tried; otherwise reassure, splint, exercises.
- **Acute dislocation:** reduce by pressing the mandible downwards and backwards (thumbs on the lower molars or external oblique ridges); advise limited opening.

9.4 Red flags in orofacial pain

- **Jaw claudication, temporal tenderness and headache over age 50:** giant cell arteritis (urgent ESR/CRP, same-day medical referral — risk to vision).
- **Progressive trismus, numbness, unexplained weight loss, lymphadenopathy, otalgia with a normal ear, nasal obstruction:**

possible malignancy (nasopharyngeal, parotid, infratemporal).

- **Cardiac pain** referred to the jaw on exertion.
- **Paroxysmal electric-shock pain triggered by light touch:** trigeminal neuralgia (carbamazepine, via GMP or specialist).

9.5 Occlusion in restorative planning

- Record and examine on mounted casts before multiple restorations; identify interferences, slide, guidance.
- **Conform to ICP** when it is stable and comfortable; **reorganise to RCP** when it is not, or when OVD is changed or extensive rehabilitation is planned.
- **Bruxism signs:** attrition facets, fractured restorations and cusps, masseteric hypertrophy, scalloped tongue, linea alba. Protect new work with a night-time hard splint; prefer metal or monolithic zirconia occlusal surfaces over layered porcelain.
- **Occlusal equilibration** (selective grinding) is irreversible: only to remove specific interferences for restorative reasons, not as TMD treatment.

Examiner traps — Unit 9

1. Selective occlusal grinding is not an evidence-based TMD treatment.
2. Reduce an acute dislocation by pressing downwards and backwards.
3. Suspected giant cell arteritis needs same-day medical referral.
4. A full-time anterior repositioning splint risks a posterior open bite.

Previous
8 · Fixed prosthodontics

Next
10 · Tooth wear and the failing dentition

TMD: diagnosis, red flags and self-care

F Pros Chapter 8 → Diploma Unit 9

DIPLOMA LEARNING OUTCOMES 2a.15, 2a.20 · 2c.19

CORE PRINCIPLE

Most common TMD presentations should be assessed conservatively; not every jaw pain is TMD.

01 Classify rather than guess

Separate muscle-related pain, TMJ arthralgia and disc/displacement disorders using the history and examination; consider the DC/TMD framework. Document pain location, movement limitation, joint noise, locking and functional impact.

02 First-line management

Explain the condition, provide reassurance, self-care, soft diet during acute symptoms, habit awareness, controlled jaw exercise and appropriate analgesia. A reversible stabilisation splint may be useful in selected cases.

03 Avoid irreversible overtreatment

Occlusal adjustment, orthodontics and extensive restorative reconstructions are not default cures for TMD. Changing an otherwise stable occlusion for pain without a sound diagnosis is a common examiner trap.

04 Recognise red flags

Suspicious swelling, unexplained neurological signs, significant trauma, systemic inflammatory features, rapidly progressive restriction or atypical pain should trigger broader assessment and appropriate referral.

05 Imaging decisions

Routine imaging is not obligatory for typical uncomplicated muscle pain. Choose targeted imaging when the clinical question concerns bone pathology, structural change or an alternative diagnosis.

SCOPE / GUIDANCE

Persistent complex TMD, diagnostic uncertainty or closed lock not responding to conservative care is a referral issue.

Source material: F Pros pp. 27–28 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 10

Tooth surface loss, the failing dentition and the perio–endo–pros interface

LO 2c.1, 2c.10–2c.12, 2c.16–2c.18, 2c.20 | Question Bank Block J (Q821–870) | Block W (Q1821–1870) | Crash notebook Schematic 22 | Number card 3

IN ONE MINUTE

Stabilise disease, prevent, then restore additively; the Dahl approach makes space without cutting tooth.

Phases: urgent → stabilisation → reassessment (50% improvement = engaging) → definitive → maintenance.

Know what is Level 2 and what is Level 3 under the NHS standard.

Decide restorability before root canal treatment.

Control disease and diagnose the cause before restoring: the NHS prosthodontic entry criteria require a stable oral environment (good hygiene, caries risk managed, active caries treated, periodontal disease stable).

10.1 Phased treatment planning

1. **Urgent care:** relief of pain and infection.
2. **Stabilisation (hygiene phase):** oral hygiene and diet advice, caries removal and temporisation, periodontal steps 1–2, extraction of hopeless teeth, transitional or interim prostheses, smoking cessation and medical risk control.
3. **Reassessment:** response to stabilisation, motivation (plaque and bleeding scores — a 50% improvement indicates an engaging patient), prognosis of remaining teeth.
4. **Definitive (corrective) phase:** endodontics, periodontal surgery if needed, then fixed and removable prosthodontics.
5. **Maintenance:** recall interval matched to risk; supportive periodontal care; prosthesis review.

10.2 Managing tooth wear

- **Identify and remove the cause** (diet diary, medical history for GORD and eating disorders, habits); refer to GMP for intrinsic causes.
- **Prevention:** reduce frequency of acidic intake and confine it to meals, drink through a straw, do not brush immediately after acid exposure, fluoride (high-concentration toothpaste, varnish), desensitising agents, night-time splint for bruxism.
- **Monitor:** photographs, study casts, silicone indices, BEWE or Smith and Knight scores.
- **Restore when** there is pain or sensitivity, aesthetic or functional concern, structural compromise or pulpal risk, or documented progression despite prevention.
- **Minimal-intervention additive approach first:** direct resin composite (repairable, adhesive, reversible), then indirect composite or gold or ceramic onlays and palatal veneers; conventional crowns last (they remove more tooth).

10.3 The Dahl concept

- **Origin:** Dahl (1975) used a removable cobalt–chromium anterior bite platform to create space for restoring worn upper anterior teeth; today the same principle uses direct composite restorations placed at an increased OVD.
- **How it works:** posterior teeth are initially out of contact. Over about 4–6 months (occasionally up to 18) occlusal contacts re-establish by a combination of intrusion of the anterior teeth and over-eruption (extrusion) of the posterior teeth, with some mandibular repositioning.
- **Best suited to** localised anterior wear where compensatory dentoalveolar growth has removed interocclusal space.
- **Relative contraindications:** active periodontal disease, implants or ankylosed teeth in the posterior segments (cannot erupt), active TMD, anterior open bite tendency, patients on drugs that affect bone turnover (bisphosphonates), poor compliance.

- **Warn patients** about initial speech change, tenderness, eating difficulty, and that posterior contact will take months.
- **Generalised wear** needing an OVD increase for multiple teeth is planned with a diagnostic wax-up and a provisional trial at the new OVD (removable splint or direct composite); extensive reorganised rehabilitation is generally Level 3.

10.4 Creating space for restorations — options

- Dahl approach (localised).
- Increase OVD (generalised; reorganised occlusion).
- Surgical crown lengthening (Level 3; risks root exposure and unfavourable crown–root ratio).
- Orthodontic intrusion or extrusion.
- Adjust opposing teeth (minor).
- Elective endodontics to gain space: **avoid**.

10.5 The failing dentition and strategic decisions

- Consider the whole mouth: strategic value of each tooth (abutment potential, occlusal stops, canine guidance), prognosis, cost, and patient priorities.
- **Shortened dental arch:** maintain anterior and premolar teeth; avoid heroic treatment of compromised molars when function is adequate.
- **Transition to edentulism:** plan stages — transitional acrylic RPDs to which teeth can be added, overdenture abutments, immediate dentures; avoid sudden total clearance where possible.
- **Retained roots under dentures:** preserve alveolar bone and proprioception; must be root-filled or asymptomatic and caries controlled.
- Teeth of doubtful prognosis should not be bridge abutments; design RPDs so teeth can be added.

10.6 Periodontal–prosthodontic interface

- **Periodontal disease must be stable** before definitive prosthodontics: PPD 4 mm or less with no BOP at those sites; full-mouth bleeding and plaque scores within indicative targets (plaque under 20%, bleeding under 30% in primary care; under 10% in Level 2–3 settings).
- **Restorations:** supragingival margins where possible, correct contour (avoid overcontour), cleansable embrasures and pontics, accurate fit.
- **RPD design for periodontally involved teeth:** tooth-supported Co-Cr with rests, open design away from gingivae, guide planes and minimal clasp torque, lingual plate for splinting mobile lower anterior teeth only if hygiene allows.
- **Splinting:** indicated when increasing mobility interferes with function or comfort; adhesive splints preferred.
- **Furcation-involved molars:** poor abutments; consider root resection only in specialist care.
- **Peri-implant mucositis** (inflammation without bone loss, reversible) is Level 2 management; **peri-implantitis** (bone loss) is Level 3.

10.7 Endodontic–prosthodontic interface

- **Restorability before root canal treatment:** sound coronal tissue, ferrule potential, periodontal support.
- **Root-filled posterior teeth:** cuspal coverage; anterior teeth: conservative unless heavily restored.
- **Bridge abutments:** assess pulp status pre-operatively; preparing vital teeth for conventional retainers carries a long-term pulpal necrosis risk (about 10–15% for bridge abutments, higher than single crowns).
- Combined complex endodontic–periodontal–prosthodontic problems on strategic teeth are Level 3.

10.8 Options for a missing tooth or teeth

OPTION	ADVANTAGES	DISADVANTAGES
Accept the space	No cost, no risk	Aesthetics, drift and over-eruption (often minimal)
Removable partial denture	Replaces several teeth and tissue, reversible, inexpensive	Removable, plaque, bulk, adaptation
Resin-bonded bridge	Minimal preparation, conservative	Debond risk, needs enamel and space
Conventional bridge	Fixed, strong, longer spans	Tooth destruction, pulpal risk, failure loses abutments too
Implant	Preserves adjacent teeth, bone stimulation, high survival	Surgery, cost, time, bone requirements, peri-implant disease; NHS access limited
Orthodontic space closure	Natural teeth	Time, appliances; Level 3 planning

10.9 Level of care (NHS prosthodontic complexity) — key boundaries

Level 1

Routine plastic, fixed and partial removable restorations that conform to the existing occlusion; removable restorations where hard and soft tissue anatomy is healthy and reasonably well formed.

Level 2

Moderately difficult patient factors: bisphosphonate, radiotherapy, haemophilia management, previous poor management, analgesia concerns. **Moderately difficult technical needs:** pre-prosthetic optimisation, occlusal reorganisation achievable with plastic restorations or an appliance, replacement and temporisation of multiple fixed units, compromised denture-bearing tissue, minor gagging, raised aesthetic expectations, simple implant-retained prostheses meeting NHS criteria.

Level 3

Major occlusal reorganisation, severe gagging or dry mouth, extensive resorption needing complete dentures, pre-prosthetic or periodontal surgery, significant TMD, implant assessment and planning, combined prosthodontic–periodontal–endodontic problems on strategic teeth, undiagnosed pain, developmental conditions (hypodontia, cleft, amelogenesis imperfecta), head and neck cancer rehabilitation.

Examiner traps — Unit 10

1. The Dahl approach fails if posterior teeth cannot erupt: implants or ankylosed teeth.
2. Periodontal stability comes before definitive prosthodontics.
3. Peri-implantitis is Level 3; mucositis is Level 2.
4. About 10–15% of conventional bridge abutments become non-vital in the long term.

Previous
9 · Occlusion and TMD

Next
11 · Implants and digital

Tooth wear: prevention before rebuilding

F Pros Chapter 9 → Diploma Unit 10

DIPLOMA LEARNING OUTCOMES 1.4, 1.8 · 2a.15, 2a.20 · 2c.10–2c.12

CORE PRINCIPLE

The exam tests the causes, progression and consequences of wear—not just how to create restorative space.

01 Identify the mechanism

Erosion is acid-mediated; attrition is tooth-to-tooth contact; abrasion is friction with an external object; abfraction is a debated stress-related model. Mixed lesions are common, so identify relevant risk factors and their frequency.

02 Establish whether to intervene

Record history, photos, charting and, where useful, BEWE or serial scans/casts. Assess dentine sensitivity, aesthetics, dento-alveolar compensation, pulp risk and whether wear is active.

03 Disease control and additive care

Manage dietary acid and reflux/eating-disorder concerns, optimise fluoride measures and monitor. For suitable localised anterior wear, Dahl-type relative axial movement can help gain space with additive restorations.

04 When rehabilitation is complex

Changes in OVD, unstable occlusion, severe generalised wear, reduced crown height or multiple failing restorations require a diagnostic wax-up, temporary testing and often specialist planning.

05 SBA traps

Do not infer reduced OVD from worn teeth alone. Do not plan full crowns before stabilising the cause of erosion or reviewing the amount of remaining tooth tissue.

SCOPE / GUIDANCE

This covers recognition and planning. Specialist crown lengthening/reorganisation is a referral pathway when indicated.

Source material: F Pros pp. 29–30 · Cross-checked against Diploma learning outcomes (Aug 2026)

Trauma: prosthodontic consequences

F Pros Chapter 12 → Diploma Unit 10

DIPLOMA LEARNING OUTCOMES 2a.7, 2a.9–2a.10, 2a.14, 2a.18 · 2c.10

CORE PRINCIPLE

F Pros includes a full trauma chapter; for the Diploma, focus on identification, urgent action, restorability and referral.

01 Emergency recognition

The source cites IADT 2020: replant an avulsed permanent tooth promptly if appropriate; never replant an avulsed primary tooth. If immediate replantation is not possible, preserve the tooth in an appropriate medium and arrange urgent expert care.

02 Longer-term complications

After trauma, monitor for pulpal necrosis, inflammatory or replacement resorption, ankylosis, colour changes and root fracture healing. Interpret radiographs and clinical signs together over time.

03 Plan around growth

An osseointegrated implant does not erupt with a growing jaw, so a fixed implant is generally deferred until growth is complete. Temporary tooth replacement or orthodontic options may preserve future choices.

04 Restorability principles

Consider sound tissue, root length, fracture position, periodontal support, ferrule and expected maintenance. A growing patient with complex traumatic injury merits multidisciplinary advice.

05 Limit of this supplement

Specific splint duration, repositioning techniques, pulpotomy and decoronation are treated in F Pros at specialist depth; they are not presented here as required Diploma operative competencies.

SCOPE / GUIDANCE

Use current IADT protocols in real emergencies and refer promptly. Trauma is contextual to the supplied Diploma outcomes.

Source material: F Pros pp. 35–36 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 11

Implant prosthodontics at Level 2 and digital workflows

LO 2a.19, 2c.5 | Question Bank Block K (Q871–920) | Block X (Q1871–1920) | Crash notebook Number card 3

IN ONE MINUTE

Primary stability is mechanical, two implants are the minimum for the edentulous mandible, and peri-implantitis needs bone loss.

Distances: 1.5 mm to a tooth, 3 mm between implants, 2 mm above the IAN.

Screw-retained crowns are retrievable; cement-retained crowns risk residual subgingival cement.

Guided surgery merges CBCT (DICOM) with surface scans (STL).

At Level 2 you are expected to provide and maintain simple implant-retained prostheses (single tooth, simple overdenture) and to diagnose peri-implant disease; implant assessment, planning and placement are Level 3.

11.1 Implant fundamentals

- **Osseointegration (Brånemark):** direct structural and functional connection between living bone and the surface of a load-bearing implant, with no intervening soft tissue.
- **Stability:** primary stability is mechanical (at placement); secondary stability is biological (bone remodelling over weeks). The "stability dip" occurs about 2–4 weeks after placement.
- **Spacing guidelines (approximate):** 1.5 mm from implant to adjacent tooth; 3 mm between implants; at least 1.5–2 mm of buccal bone; 2 mm safety margin above the inferior alveolar canal.
- **Loading protocols (ITI):** immediate (within 1 week), early (1 week to 2 months), conventional (after 2 months).
- **Connections:** external hex (historic, more screw loosening), internal hex or octagon, conical (Morse taper) — better seal and stability. Platform switching (narrower abutment than implant) may reduce crestal bone loss.
- **Risk factors for failure and peri-implantitis:** smoking, poorly controlled diabetes, history of periodontitis, poor plaque control, lack of maintenance, radiotherapy to the site, high-dose IV antiresorptives, bruxism.

11.2 Implant prostheses

OPTION	KEY POINTS
Two-implant mandibular overdenture	McGill (2002) and York (2009) consensus minimum standard for the edentulous mandible; big gains in satisfaction and function
Unsplinted attachments (ball, locator-type)	Easier hygiene and maintenance, less restorative space; matrix (insert) wear needs replacement
Bar-retained overdenture	Splints implants, more retention, useful with divergent implants or in the maxilla; needs more vertical space and hygiene effort
Maxillary implant overdenture	Usually 4 or more implants; palateless design possible
Single implant crown	Screw-retained (retrievable, no subgingival cement) or cement-retained (passive, aesthetic access, but residual cement causes peri-implant disease)
Fixed full-arch	Level 3; passive fit essential; distal cantilevers kept short relative to implant spread

- **Most common complications:** overdentures — loss of retention and attachment insert replacement, denture fracture opposite implants; single crowns — abutment screw loosening, ceramic chipping; fixed full-arch — veneer chipping and fracture.
- Use a calibrated torque driver to the manufacturer's value for abutment and prosthetic screws.

- **Implants have no periodontal ligament:** no physiological movement or proprioception (osseoperception only). In a tooth–implant connected bridge the differential mobility causes problems: avoid.

11.3 Peri-implant health and disease (2017 World Workshop)

CONDITION	FINDINGS	MANAGEMENT LEVEL
Peri-implant health	No erythema, bleeding or suppuration; no bone loss beyond initial remodelling	Maintenance
Peri-implant mucositis	Bleeding and/or suppuration on gentle probing; no bone loss beyond initial remodelling; reversible	Level 2: mechanical debridement, OHI, check prosthesis design, adjuncts
Peri-implantitis	Bleeding and/or suppuration, increased probing depth, progressive bone loss. Without baseline: bone level 3 mm or more apical to the implant platform and PPD 6 mm or more with bleeding	Level 3 (when NHS responsibility)

- Record baseline radiograph and probing depths at the time of prosthesis fitting.
- **Maintenance:** patient-performed cleaning (interdental brushes, floss under bars), professional debridement with instruments that do not scratch titanium, annual review.

11.4 Digital prosthodontics

- **Intraoral scanners:** excellent for short spans and single units; accuracy (trueness and precision) falls over full arches due to stitching errors. Limitations: subgingival margins, blood and saliva, edentulous mucosa (cannot mucodisplace or border mould).
- **File formats:** STL (surface triangles, open format, no colour), PLY or OBJ (can carry colour and texture), DICOM (CBCT and medical imaging). Merging DICOM with STL enables guided implant surgery.
- **Subtractive manufacture (milling):** dense, homogeneous materials (pre-polymerised PMMA, zirconia, lithium disilicate, PEEK, Co-Cr, Ti); limited by bur size for fine detail.
- **Additive manufacture (3D printing):** stereolithography (SLA), digital light processing (DLP), selective laser melting (metals). Uses: models, surgical guides, trays, splints, temporary crowns, printed denture bases and teeth.
- **Digital complete dentures:** fewer visits, records stored for easy duplication, milled bases have low residual monomer and excellent fit; limitations in recording functional borders and in complex jaw relations.
- **Digital smile design** and virtual mock-ups help communication and consent; a physical mock-up from a wax-up is still useful for patient approval.
- Virtual articulators and facial scanning integrate facial references; jaw motion tracking can record dynamic occlusion.

Examiner traps — Unit 11

1. The stability dip falls at 2–4 weeks, as primary gives way to secondary stability.
2. Avoid rigidly connecting a tooth to an implant.
3. Intraoral scans are least accurate on mobile edentulous mucosa and long spans.
4. Clean titanium with instruments that do not scratch it.

Previous
10 · Tooth wear and the failing dentition

Next
12 · Complications and maintenance

Implants: planning, disease and maintenance

F Pros Chapter 7 → Diploma Unit 11

DIPLOMA LEARNING OUTCOMES 2a.19 · 2c.2, 2c.4–2c.5, 2c.21

CORE PRINCIPLE

The Diploma emphasis is prosthodontic choices, recognition of disease and safe referral—not advanced implant surgery.

01 Prosthesis-first assessment

Identify the missing-tooth problem, desired prosthesis, smile line, available restorative space, adjacent teeth, occlusion, cleansability and patient priorities before considering implant feasibility.

02 Restorative design choices

Discuss the implications of screw-retained versus cement-retained prostheses. Screw access and retrievability can be advantages; excess subgingival cement is a recognised peri-implant inflammatory risk.

03 Detect complications

Examine peri-implant tissues, bleeding or suppuration, plaque control, restoration integrity and radiographic bone levels in context. Distinguish peri-implant mucositis from suspected peri-implantitis, document findings and arrange appropriate review/referral.

04 Long-term maintenance

Outline hygiene, supportive care, prosthesis access and implant-specific complications such as screw loosening, chipping and biological inflammation. An implant surviving in situ is not synonymous with a successful functioning restoration.

05 Do not overextend the scope

Sinus grafting, ridge augmentation, complex surgical templates, advanced full-arch implants and specialist peri-implant surgery appear in F Pros but should not be treated as routine Level 2 Diploma procedures.

SCOPE / GUIDANCE

Source NHS complexity chart directs complex implant planning and multidisciplinary implant-related care to Level 3.

Source material: F Pros pp. 24–26 · Cross-checked against Diploma learning outcomes (Aug 2026)

Digital workflows: strengths and limitations

F Pros Chapter 13 → Diploma Unit 11

DIPLOMA LEARNING OUTCOMES 1.9 · 2c.5 · 3a.5

CORE PRINCIPLE

Digital is a method—not proof that the impression, plan or result is accurate.

01 When an intraoral scan helps

Good access and controlled moisture permit rapid review and rescanning, especially for single units and shorter spans. Accuracy depends on visibility, scanner technique, software and surface characteristics.

02 When conventional methods remain useful

Full-arch stitching, movable edentulous tissues, functional impressions, subgingival bleeding margins and multiple-implant passive fit can create limitations. Use a validated technique and verification where indicated.

03 Know the file vocabulary

Trueness means closeness to the reference; precision means repeatability. STL contains surface geometry; DICOM is used for volumetric imaging. Merging scans can support prosthetically led planning, but mismatches must be checked.

04 CAD/CAM options

Milling provides reproducible dense industrial materials but has tool-radius and material-waste limits. Printing is useful for models, trays and provisionals; post-curing and material indication affect outcomes.

05 Records and consent

Use digital design to explain options, not to promise a perfect smile. Save source files securely, document revisions, and check the limits and validation status of the individual device/workflow.

SCOPE / GUIDANCE

Complex guided implant surgery and advanced computer-assisted rehabilitation belong under appropriately trained services.

Source material: F Pros pp. 37–38 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 12

Complications, failures and maintenance

LO 2c.21–2c.27 | Question Bank Block L (Q921–950) | Block Y (Q1921–1950) | Crash notebook Schematic 27

IN ONE MINUTE

Find the cause before you replace: most failures are retention, caries or fatigue.

Repeated decementation → resistance form. **Porcelain fracture** → unsupported porcelain.

A framework that will not seat → a distorted impression or cast.

Survival (still in place) is not **success** (free of complications).

When a restoration fails, the exam wants you to find the cause before replacing it: an identical replacement will fail the same way.

12.1 Fixed restorations

PROBLEM	LIKELY CAUSES	MANAGEMENT
Crown or bridge decementation	Over-tapered or short preparation, inadequate resistance form, caries under the margin, parafunction, long span or pier abutment, poor cementation	Check caries and preparation; recement only if sound and retentive; improve retention (grooves, adhesive cement) or redesign
Unnoticed loosening of one retainer (fixed-fixed)	Differential movement; bridge still held by the other retainer	Remove the bridge; treat caries; redesign (cantilever)
Porcelain chipping or fracture	Inadequate metal support, porcelain over 2 mm unsupported, occlusal overload, bruxism, sharp internal angles, CTE mismatch	Small chips: polish or composite repair (HF + silane on ceramic, MDP on exposed metal); larger: replace and redesign with metal or zirconia occlusal surfaces
Secondary caries	Poor margins, overhangs, high caries risk, open contacts	Prevention; repair if accessible; replace
Post-operative sensitivity	Dentine exposure, microleakage, thermal trauma during preparation, high occlusion	Check occlusion; desensitise; monitor sensibility
Pulpal necrosis or apical periodontitis	Cumulative trauma (caries, preparation, impressions, temporaries)	Endodontic treatment through the crown and seal the access, or replace
Root fracture under a post	Tapered or oversized post, no ferrule, excessive loading	Usually extraction; prevention through ferrule and post design
Gingival inflammation or recession	Overcontour, subgingival margins invading the supracrestal attachment, rough margins, excess cement	Correct contour; remove excess cement; replace if margins defective
Food packing	Open or weak contact, marginal ridge discrepancy, plunger cusp	Adjust or replace the restoration
Resin-bonded bridge debond	Inadequate wing coverage or enamel, occlusal interference, contamination at bonding	Clean, re-sandblast, rebond under rubber dam with MDP cement; check occlusion and redesign if repeated

Literature anchors: for conventional bridges, caries and loss of retention are the leading causes of failure; the need for endodontic treatment is a common biological complication of crowned vital teeth.

12.2 Removable partial dentures

PROBLEM	LIKELY CAUSES
Framework does not seat at fit	Distorted impression or cast, tray movement, rest seats not recorded, investment or casting error, soft tissue impingement — check with disclosing medium; if gross, re-impression and remake
Clasp loose / lost retention	Clasp distorted (patient removing by the clasp), work hardening, wrong undercut, wear
Clasp fracture	Fatigue at the shoulder, repeated adjustment of Co-Cr (low ductility), porosity in casting
Abutment pain or mobility	Occlusal overload, rest not seated, free-end saddle rotation (needs reline), clasp active at rest
Free-end saddle sinks and indirect retainer lifts	Ridge resorption — reline the saddle
Gingival inflammation under the denture	Gingival coverage, plaque, inadequate relief, poor hygiene
Caries of abutments	Plaque retention at clasps and minor connectors, high sugar intake
Food under saddles	Under-extension, poor saddle fit

12.3 Complete dentures — maintenance

- **Recall:** at least annually for mucosal examination, fit and occlusal wear.
- **Reline or rebase** when fit is lost but the polished and occlusal surfaces are acceptable; **remake** when OVD, aesthetics, or occlusion are also wrong.
- **Repairs:** fracture lines must be reassembled accurately before a cast is poured; self-cured acrylic repairs are weaker; investigate the cause of repeated fractures (Section 7.9).
- **Denture hygiene:** brush with a soft brush and soap or non-abrasive cleanser over a basin of water, soak overnight (hypochlorite only for acrylic, no metal), leave out at night.

12.4 Laboratory and external factors

- **Errors arise at:** impressions (distortion, drag, voids), casts (air bubbles, abraded dies), articulation (incorrect records), wax and investment (distortion), casting (porosity, incomplete casting), processing (acrylic porosity, raised OVD from flash), and finishing (over-polished fitting surfaces).
- **Prevention:** clear written prescriptions, quality control at each clinical stage (inspect impressions and casts on return), good communication including photographs and shade, agreed delivery times.

12.5 Assessing outcomes

- Compare outcomes with published survival data and with the patient's goals (patient-reported outcomes: comfort, function, aesthetics, satisfaction).
- **Survival** = the prosthesis is still in situ; **success** = in situ and free of complications. Ask which is reported when interpreting studies.

PROSTHESIS	APPROXIMATE BENCHMARK (META-ANALYSES)
Conventional fixed bridges	About 89% survival at 10 years
Single metal–ceramic crowns	About 95% at 5 years
Resin-bonded bridges (modern)	About 80–90% at 5–10 years, with cantilevers performing well

Implants	About 95% survival at 10 years
Implant overdenture attachments	Need frequent maintenance

Examiner traps — Unit 12

1. Indirect retainers lifting when the saddle is pressed means the saddle needs relining.
2. Bleeding and swelling round one new crown is residual cement until proven otherwise.
3. The laboratory prescription is a legal document; laboratories must be MHRA-registered and supply a statement of manufacture.
4. An intact debonded resin-bonded bridge with no caries is cleaned and rebonded, not remade.

Previous
11 · Implants and digital

Next
13 · Governance, law and communication

Failure analysis and measurable success

F Pros Chapters 5–7 and 11 → Diploma Unit 12

DIPLOMA LEARNING OUTCOMES 2c.21–2c.27 · 3a.2–3a.8

CORE PRINCIPLE

A sound answer diagnoses why a prosthesis failed before choosing the repair or remake.

01 Diagnose by category

Biological: new caries, pulpal symptoms, periodontal/peri-implant inflammation. Mechanical: fractured tooth/restoration, clasp failure, screw loosening or debond. Functional: occlusal interference, instability or inadequate support.

02 Review the sequence of failure

Ask when symptoms began, whether fit/occlusion altered, whether maintenance occurred and whether risk factors changed. Use history, clinical examination, photographs, tests, radiographs and old records as justified.

03 Correct cause—not appearance

An RPD repeatedly fracturing may reflect inadequate thickness, support or occlusal forces. A debonding crown may signal poor retention form. Excess cement, inaccessible hygiene and design can underlie peri-implant inflammation.

04 Audit outcomes

Document comfort, function, aesthetics, cleaning, biological health, mechanical integrity and patient satisfaction. Survival percentages alone do not capture complications or burden of maintenance.

05 Feedback to the plan

Treat failure as an opportunity to change the design, risk control, consent or review interval. Escalate repeated complex failures or combined specialist problems beyond Level 2 competence.

SCOPE / GUIDANCE

Case-based reasoning and honest complication communication are more important than memorising unsupported survival figures.

Source material: F Pros pp. 20, 22–26, 33–34 · Cross-checked against Diploma learning outcomes (Aug 2026)

UNIT 13

Maintaining good practice: governance, law and communication

LO 3a.1–3a.12, 3b.1–3b.5 | Question Bank Block M (Q951–1000) | Block Z (Q1951–2000) | Crash notebook Schematics 25–26 | Number card 4

IN ONE MINUTE

Montgomery for consent, the MCA for capacity, candour when things go wrong, and adult records kept 11 years.

GDC: nine principles; enhanced CPD 100 hours in 5 years.

IR(ME)R duty holders and **HTM 01-05** decontamination.

Safeguarding: record and refer; never investigate yourself.

These are the most learnable marks in the paper: the answers follow UK law and GDC standards, not opinion. The rules below are for England unless stated; check devolved-nation equivalents.

13.1 Consent and capacity

- **Valid consent** is informed, voluntary, given by a person with capacity, and specific to the treatment; it is a process, documented in the notes, and can be withdrawn at any time.
- **Montgomery v Lanarkshire (2015):** the clinician must disclose material risks — those a reasonable person in the patient's position would consider significant, or that this particular patient would — and reasonable alternatives (including no treatment). It replaced Bolam as the test for disclosure.
- **Bolam (1957):** standard of care is that of a responsible body of professional opinion; **Bolitho (1997):** that opinion must withstand logical analysis. Both still apply to diagnosis and treatment (not disclosure).
- **Negligence** requires duty of care, breach of that duty, causation, and resulting harm. **Vicarious liability:** an employer is liable for the acts of employees in the course of employment.
- **Mental Capacity Act 2005** (England and Wales) principles: presume capacity; take all practicable steps to help the person decide; an unwise decision does not mean incapacity; act in best interests; choose the least restrictive option. **Two-stage test:** impairment of mind or brain, then can the person understand, retain, use or weigh, and communicate the decision? Capacity is decision-specific and time-specific.
- **Scotland:** Adults with Incapacity (Scotland) Act 2000.
- **Young people:** 16–17-year-olds are presumed to have capacity to consent (Family Law Reform Act 1969); under 16s may consent if Gillick competent.
- **Shared decision-making (NICE NG197):** discuss benefits, risks, alternatives and doing nothing (BRAN); use decision aids; check understanding.

13.2 GDC Standards for the Dental Team — nine principles

1. Put patients' interests first.
2. Communicate effectively with patients.
3. Obtain valid consent.
4. Maintain and protect patients' information.
5. Have a clear and effective complaints procedure.
6. Work with colleagues in a way that is in patients' best interests.
7. Maintain, develop and work within your professional knowledge and skills.
8. Raise concerns if patients are at risk.
9. Make sure your personal behaviour maintains patients' confidence in you and the dental profession.

- **Enhanced CPD (dentists):** at least 100 verifiable hours per 5-year cycle, at least 10 hours in every 2-year period, with a personal development plan; recommended topics include medical emergencies, decontamination, radiography and radiation protection, legal and ethical issues, complaints handling, oral cancer detection, safeguarding.
- Appropriate indemnity is mandatory.

13.3 Duty of candour, incidents and complaints

- **Professional duty of candour** (all registrants): be open and honest when something goes wrong, apologise, offer an appropriate remedy, explain the short- and long-term effects. **Statutory duty of candour** applies to CQC-registered providers for notifiable safety incidents. An apology is not an admission of liability.
- **Incident reporting:** significant event analysis and root cause analysis locally; NHS patient safety events to the national system (Learn from Patient Safety Events); adverse drug reactions via MHRA Yellow Card; medical device incidents to MHRA; serious workplace injuries and dangerous occurrences under RIDDOR to the HSE.
- **Never events** are wholly preventable serious incidents (e.g., wrong-site surgery including wrong tooth extraction).
- **NHS complaints (England):** acknowledge within 3 working days; resolve locally; complaint normally made within 12 months of the event or of becoming aware; escalation to the Parliamentary and Health Service Ombudsman. **Private patients:** the Dental Complaints Service. Respond in writing, apologise where appropriate, explain, and describe actions taken.
- **Raising concerns (whistleblowing)** is protected by the Public Interest Disclosure Act 1998; GDC requires registrants to raise concerns even if they are not the employer.

13.4 Records and information governance

- Records must be accurate, contemporaneous, legible, attributable, and never altered without a dated, signed note of the change.
- **Retention (NHS England guidance):** adults at least 11 years after the last treatment; children until age 25 or 11 years after the last treatment, whichever is longer. Radiographs and models are part of the record.
- **UK GDPR and the Data Protection Act 2018:** lawful basis for processing, security, data minimisation. Subject access requests answered within one month, normally free of charge. Caldicott principles govern confidential information sharing.
- **Confidentiality may be breached** with consent, when required by law (court order), or in the public interest (serious harm to others) — share the minimum necessary.

13.5 Radiation protection

- **IRR17 (Ionising Radiations Regulations 2017):** protect workers and public; employer registers with the HSE, appoints a radiation protection adviser (RPA) and radiation protection supervisor (RPS), sets local rules and a controlled area (for intraoral sets, typically within 1.5 m of the tube and patient and within the primary beam until attenuated).
- **IR(ME)R 2017:** protect patients; duty holders are employer, referrer, practitioner (justifies) and operator; justification, optimisation (ALARP), clinical evaluation and recording of every exposure; pregnancy enquiries where relevant.
- **Optimisation tools:** rectangular collimation, film holders, fastest image receptor, appropriate kV and exposure, quality assurance programme and audit of image quality.

13.6 Infection prevention and control (HTM 01-05)

Decontamination cycle

Transport → Cleaning → Inspection → Sterilisation → Storage → Use

- **Cleaning:** washer-disinfector is best practice; ultrasonic or manual acceptable for essential quality. **Inspection** under illuminated magnification.
- **Steam sterilisation** at 134–137°C for at least 3 minutes. Vacuum (type B) sterilisers are needed for wrapped and hollow or lumened instruments; non-vacuum (type N) for unwrapped solid instruments.
- Daily automatic control test (and steam penetration test for vacuum sterilisers); periodic testing and validation by a competent

person.

- Separate dirty-to-clean workflow; ideally a dedicated decontamination room.
- **Prosthodontic items:** rinse and disinfect impressions, records, try-ins and appliances before they go to or come back from the laboratory; label as disinfected.
- **Dental unit waterlines:** flush at the start of the day (about 2 minutes) and between patients (20–30 seconds); use biocide; test microbial quality.
- **Sharps injury:** encourage bleeding, wash with soap and water (no scrubbing), cover, report, contact occupational health urgently (HIV post-exposure prophylaxis ideally within 1 hour, up to 72 hours); assess source patient risk with consent.
- Hepatitis B vaccination with documented response for exposure-prone procedures.

13.7 Safeguarding, equality and diversity

- **Safeguarding children and adults at risk:** recognise (injuries to head, face and mouth; dental neglect; unusual behaviour), record, refer (practice safeguarding lead, local authority children's or adult social care; police if immediate danger). Do not investigate yourself; information sharing to protect a child or adult is lawful.
- **Female genital mutilation:** regulated health professionals in England and Wales must report known cases in under-18s to the police.
- **Equality Act 2010:** nine protected characteristics (age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, sexual orientation); duty to make reasonable adjustments (e.g., wheelchair access, longer appointments, easy-read information).

13.8 Clinical governance, audit and evidence

- **Pillars:** clinical effectiveness and audit, risk management, education and training, patient and public involvement, information management, staff management, research.

Audit cycle

Choose topic → Set standard → Measure current practice → Compare with standard → Implement change → Re-audit (closes the loop)

- **Audit** measures practice against a standard; **research** generates new knowledge; **service evaluation** describes current service without a standard.
- **Quality improvement:** Plan–Do–Study–Act cycles, run charts, root cause analysis, significant event analysis, human factors, checklists (e.g., local safety standards for invasive procedures).
- **Hierarchy of evidence:** systematic review and meta-analysis of RCTs → RCT → cohort → case–control → case series and reports → expert opinion. GRADE rates certainty of evidence.
- **Key statistics:** sensitivity (true positive rate) and specificity (true negative rate); positive predictive value depends on prevalence; number needed to treat = 1 ÷ absolute risk reduction; a 95% confidence interval crossing 1 (for ratios) or 0 (for differences) is not statistically significant.
- **Regulators of services:** CQC (England), Healthcare Improvement Scotland, Healthcare Inspectorate Wales, RQIA (Northern Ireland). **Professional regulator:** GDC.
- **Level 2 practitioners** must have a formal link with a consultant-led managed clinical network, complete a minimum caseload, record outcomes (PROMs and PREMs), audit, and report complications.

13.9 Communication

- **Breaking bad news (SPIKES):** Setting, Perception (what do they know?), Invitation (how much do they want to know?), Knowledge (warning shot, plain language, small chunks), Emotions (acknowledge and empathise), Strategy and Summary.
- **Language barriers:** professional interpreter, not family members or children.
- **Hearing impairment:** face the patient, remove mask when safe, written information, BSL interpreter. **Learning disability:** easy-read and Makaton, involve carers, longer appointments. **Dementia:** short simple choices, consistent clinician and environment.

- **Children:** age-appropriate language, tell–show–do, involve parents.
- **Referral letters:** patient details, reason for referral and question to answer, relevant history, medications, findings, radiographs, urgency, patient's wishes; copy to the patient where appropriate.
- **Written treatment plans and estimates** (NHS FP17DC in England) support informed consent and reduce complaints.

Examiner traps — Unit 13

1. An apology is not an admission of liability.
2. 16- and 17-year-olds are presumed to have capacity; under-16s need Gillick competence.
3. Audit measures practice against a standard; research creates new knowledge.
4. Extracting the wrong tooth is a never event.

Previous
12 · Complications and maintenance

Next
14 · High-yield numbers

UNIT 14 · FINAL WEEK

High-yield numbers and last-minute facts

Read this in the final week; every row has appeared, or is likely to appear, as a single-fact SBA. Grouped by topic — the unit link on each card takes you back to the full notes.

The exam 150 SBAs · 180 minutes · no negative marking

Complete dentures Unit 7

Freeway space	2–4 mm (average about 3 mm)
Closest speaking space ("s")	About 1–2 mm
Upper incisors vs incisive papilla	Labial surface about 8–10 mm anterior
Lower occlusal plane	About two-thirds up the retromolar pad
Post-dam depth	About 1–1.5 mm, shallower over the midline raphe

Partial dentures Unit 6

Lingual bar space	At least 7–8 mm from gingival margin to functional floor of mouth
Mid-palatal strap width	At least 8 mm
Maxillary connector clearance from gingival margin	About 6 mm (mandible about 3 mm) if not covering
Clasp undercut	Co-Cr 0.25 mm; cast gold 0.5 mm; wrought wire 0.5–0.75 mm
Co-Cr clasp length	At least about 15 mm to engage 0.25 mm
Clasp / span flexibility	Proportional to length cubed
Occlusal rest seat depth	At least 1–1.5 mm (Co-Cr)
Guide plane height	2–4 mm

Materials Unit 3

Elastic modulus	Co-Cr about 220 GPa; Type IV gold about 90–100 GPa; Ti about 110 GPa; zirconia about 210 GPa; enamel about 85 GPa; dentine about 18 GPa; PMMA about 2.5 GPa
PMMA shrinkage	Monomer 21% volume; dough about 7% volume, 0.5% linear
MMA boiling point	100.3°C (gaseous porosity)
Residual monomer	Heat-cured about 0.2–0.5%; self-cured about 2–5%
Gypsum W/P	Plaster 0.45–0.5; stone 0.28–0.30; die stone 0.22–0.24
Luting cement film thickness (ISO)	25 µm maximum
Zirconia volume increase on transformation	About 3–5%
Zirconia sintering shrinkage	About 20–25% linear
Lithium disilicate etch	5% HF, about 20 s
Enamel etch	30–37% phosphoric acid, 15–30 s

Fixed prosthodontics Unit 8

Taper (total convergence)	Ideal 6°; clinically 10–20°
Minimum crown preparation height	3 mm anterior and premolar; 4 mm molar
Metal–ceramic labial reduction	1.2–1.5 mm
Veneer preparation	0.3–0.7 mm, within enamel
Ferrule	1.5–2 mm height, about 1 mm thickness
Post	Length at least crown height or two-thirds root; leave 4–5 mm GP; diameter at most one-third root width
Supracrestal tissue attachment	About 2 mm; keep margins about 3 mm from bone crest
Crown–root ratio	Ideal 2:3; minimum 1:1
Hygienic pontic clearance	At least 3 mm
Lithium disilicate bridge connector	12 mm² anterior, 16 mm² premolar; 3 units max
Zirconia bridge	About 9 mm² (3-unit posterior)

connector	

Implants Unit 11

Implant distances	1.5 mm to tooth; 3 mm between implants; 2 mm above IAN canal
Implant loading	Immediate under 1 week; early 1 week–2 months; conventional over 2 months
Peri-implantitis (no baseline)	Bone 3 mm or more below platform plus PPD 6 mm or more with bleeding
McGill / York consensus	Two-implant mandibular overdenture as minimum standard

Perio, wear and physiology Units 2, 5, 10

Critical pH	Enamel 5.5; root dentine about 6.2–6.7; fluorapatite about 4.5
BSP grade	Bone loss %/age: under 0.5 = A; 0.5–1.0 = B; over 1.0 = C
BSP stable	BOP under 10%, PPD 4 mm or less, no BOP at 4 mm sites
Patient engagement	50% or more improvement in plaque and bleeding; plaque under 20%, bleeding under 30%
BEWE	Per sextant 0–3; cumulative 14 or more = high risk
Hyposalivation	Unstimulated under 0.1 mL/min
Mouth opening	Normal about 40–55 mm; under 40 mm restricted
Dahl	Posterior contacts re-establish in about 4–6 months
Shortened dental arch	3–5 occlusal units adequate (Käyser)
WHO	At least 20 functional teeth for life

Medicine and emergencies Unit 4

HbA1c diabetes	48 mmol/mol (6.5%) or more
Warfarin	Treat in primary care if INR under 4 (check within 24 hours)
Anaphylaxis adult	Adrenaline 1:1000 IM 500 micrograms (0.5 mL)
Glucagon	1 mg IM
GTN	400 micrograms sublingual
Buccal midazolam	10 mg adult
Aspirin (MI)	300 mg chewed
Inhalation sedation	At least 30% oxygen (maximum 70% N_2O)
MRONJ	Exposed bone more than 8 weeks
Ulcer referral	Over 3 weeks unexplained → urgent suspected cancer referral

Governance Unit 13

Steam sterilisation	134–137°C, at least 3 minutes
Records (adult)	At least 11 years after last treatment
Subject access request	One month
NHS complaint acknowledgement	3 working days; complaint within 12 months
CPD (dentists)	100 hours per 5 years; at least 10 hours every 2 years

Previous
13 · Governance, law and communication

Back to
Contents

Final-week integrated decision traps

F Pros crash notebook → Diploma Unit 14

DIPLOMA LEARNING OUTCOMES Across 1, 2a, 2b, 2c and 3a–3b

CORE PRINCIPLE

Use these prompts to test the decisions common to both source notebooks—not as official examination predictions.

01 Diagnose before intervening

A suspicious lesion needs investigation/referral, not camouflage by a prosthesis. In ambiguous test results, repeat or complement the clinical examination instead of forcing a diagnosis.

02 Stabilise and preserve

Caries/periodontal control precedes complex definitive work. Prefer additive or reversible approaches where they meet the need; no treatment is a legitimate comparator for a missing tooth.

03 Design with forces in mind

RPD support precedes clasp selection; a distal extension needs indirect retention and maintenance. Low denture retention can be functional-border failure rather than an adhesive deficiency.

04 Material chemistry matters

A post retains, not reinforces. Silane is for suitable silica-based surfaces, not zirconia. Good cement cannot rescue a short and severely over-tapered preparation.

05 Recognise your level

Routine Level 2 planning may include moderately difficult tooth replacement; complex multidisciplinary care, severe parafunction, full-mouth reorganisation, complex implants and specialist trauma management need escalation.

06 Answer the clinical question asked

Look for “first,” “most appropriate” or “least invasive,” and select the patient-safe next action rather than the most sophisticated possible procedure.

SCOPE / GUIDANCE

Revision prompts are educational summaries of the supplied notes; exact standards and medications require current verification.

Source material: F Pros pp. 5–7, 10, 12, 15, 17, 20, 23, 26, 28, 30, 32, 34, 36, 38 · Cross-checked against Diploma learning outcomes (Aug 2026)